

Zero-overhead deterministic exceptions: Throwing values

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Abstract

Divergent error handling has fractured the C++ community into incompatible dialects:

- (1) C++ projects often ban even turning on compiler support for exception handling, but this means they are not using Standard C++. Exceptions are required to use central C++ standard language features (e.g., constructors) and the C++ standard library. Yet in [\[SC++F 2018\]](#), **over half** of C++ developers report that exceptions are banned in part (32%) or all (20%) of their code, which means they are using a divergent language dialect with different idioms (e.g., two-phase construction) and either a nonconforming standard library dialect or none at all. We must make it possible for all C++ projects to at least turn on exception handling support and use the standard language and library.
- (2) We keep inventing more incompatible error handling mechanisms, including in the standard library. We should support common proven ones in [try/throw/catch](#) so they do not need to be divergent libraries.

This paper aims to extend C++’s exception model to let functions declare that they **throw a statically specified type by value**. This lets the exception handling implementation be exactly as efficient and deterministic as a local return by value, with zero dynamic or non-local overheads.

The proposed extension follows modern existing practice (e.g., `std::error_code`, Expected [\[P0323R3\]](#), Outcome [\[Douglas 2018\]](#)), and it doubles down on C++’s core strength of efficient value semantics, just as when we added move semantics as a C++11 marquee feature.

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1 Overview

1.1 Prelude: Success-with-info and bugs/corruption are not “errors”

error: “an act that ... fails to achieve what should be done.” — [Merriam-Webster]
“throwing logic error type exceptions... is not recoverable...” — [Douglas 2018]

In this paper, “error” means exactly and only “a function couldn’t do what it advertised” — its preconditions were met, but it could not achieve its successful-`return` postconditions, and the calling code can recover.

(1) An alternate result is never an “error” (it is success, so report it using `return`). This includes “partial success” such as that a buffer was too small for the entire request but was filled to capacity so more can be read on the next call. Consider opening a file: For a general `disk_file` class’s constructor that takes a filename, not being able to open the file is a normal outcome (due to file system race conditions) and the type should allow a not-open state; failing to open the file is *not an error*, it does not violate the constructor’s postcondition (its class invariant). For an `open_file` class whose invariant guarantees an object of that type represents an *opened* file, or for a higher-level function like `InitializeSubsystem` that depends on a `config.ini` file, not being able to open the file is an error because it prevents achieving that constructor’s postcondition (its class invariant).

Note I distinguish “error codes” from “status codes” that could contain non-error results. Conflating “routine” and “error” results in one object makes it hard to keep normal and error processing distinct.

(2) A programming bug or abstract machine corruption is never an “error” (both are not programmatically recoverable, so report bugs to the human programmer using contract violations that default to fail-fast and report abstract machine corruption using fail-fast). Programming bugs (e.g., out-of-bounds access, null dereference) and abstract machine corruption cause a corrupted state that cannot be recovered from programmatically, and so should never be reported to the calling code as errors that code could somehow handle. For example, these are *not errors*:

- A precondition (e.g., `[[expects...]]`) violation is always a bug in the caller (it shouldn’t make the call). Corollary: `std::logic_error` and its derivatives should never be thrown (§4.2); use contracts instead.
- A postcondition (e.g., `[[ensures...]]`) violation on “success” return is always a bug in the callee (it shouldn’t return success). Violating a `noexcept` declaration is also a form of postcondition violation.
- An assertion (e.g., `[[assert...]]`) failure is always a bug in the function (its code is incorrect).
- Stack exhaustion is always an abstract machine corruption (a function cannot guard against it).

Note For a discussion of heap exhaustion (OOM), see §4.3.

1.2 Background and motivation summary

“... error handling idioms and practices remain contentious and confusing within the C++ community (as well as within the Boost community).” — [Bay 2018]

C++ is the only major language without a uniform error handling mechanism that is recommendable for all code:

- (§2.1) Neither today’s exceptions nor error codes are it. Each meets requirements that the other does not.
- (§2.2) We are proliferating dual-mode interfaces that try to support *both* models (e.g., `std::filesystem`).
- (§2.3) Worse, for reasonably common classes of real-world examples, *neither* is acceptable.
- (§2.4) So we keep trying to invent new alternatives (e.g., Expected, Outcome).

One consequence is that “C++” projects commonly turn off exception handling support in all or part of their projects, and are therefore not actually using Standard C++, but using a divergent language dialect with different idioms (e.g., two-phase construction) and either a nonconforming standard library dialect or none at all.

We must make it possible for all C++ projects to at least turn on exception handling support so they can use the standard language and library. So the question is not how to pick a winner from among the many error handling mechanisms; it is how to improve `throw/try/catch`, the only alternative that supports basic features like constructors, to become a universally usable mechanism, given C++’s constraints of zero-overhead and determinism (both of which today’s dynamic exception model violates) and backward source and binary compatibility.

Note Importantly, “zero overhead” is not claiming zero *cost* — of course using something always incurs some cost. Rather, C++’s zero-overhead principle has always meant that (a) “you don’t pay for what you don’t use” and (b) “when you do use it you can’t [reasonably] write it more efficiently by hand.”

1.3 Design principles

Note These principles apply to all design efforts and aren’t specific to this paper. Please steal and reuse.

The primary design goal is conceptual integrity [Brooks 1975], which means that the design is coherent and reliably does what the user expects it to do. Conceptual integrity’s major supporting principles are:

- **Be consistent:** Don’t make similar things different, including in spelling, behavior, or capability. Don’t make different things appear similar when they have different behavior or capability. — This proposal preserves the clear distinction between normal code and error handling, both when raising an error (`throw` instead of `return`) and handling an error (`catch` instead of normal control flow like `if/co_await`). It aims to remove the incentive to resort to compromised designs such as having the same function inconsistently report some errors using an error code and others by throwing an exception. It directly improves consistency by providing a form of exception whose automatic propagation leaks much less type information from otherwise-encapsulated implementation details.
- **Be orthogonal:** Avoid arbitrary coupling. Let features be used freely in combination. — This proposal enables better composition by making errors that are propagated across a boundary cleanly composable, including supporting better automatic lossless propagation.
- **Be general:** Don’t restrict what is inherent. Don’t arbitrarily restrict a complete set of uses. Avoid special cases and partial features. — This proposal retains the ability to report all kinds of errors using the same mechanism (with the note that heap exhaustion may be worth treating specially; see §4.3). It aims to evolve exception handling to be a single general way to report errors that is suitable for all C++ code.

These also help satisfy the principles of least surprise and of including only what is essential, and result in features that are additive and so directly minimize concept count (and therefore also redundancy and clutter).

1.4 Acknowledgments

Thank you especially to the Direction Group for publishing [P0939R0] as a call for a direction for C++ evolution that explicitly calls out the need to address the ongoing practical problems with exceptions vs. error codes.

Thank you to SG14 for their feedback, and to the following for reviews and comments on this material: JF Bastien, Charley Bay, Paul Bendixen, Vicente Botet, Jonathan Caves, Alex Christensen, Ben Craig, Pavel Curtis, Guy Davidson, Gabriel Dos Reis, Niall Douglas, Chris Guzak, Howard Hinnant, Odin Holmes, Andrew Hunter, Dan Jump, Ben Kuhn, Stephan T. Lavavej, Jason Merrill, Phil Nash, Gor Nishanov, Michael Novak, Arthur O’Dwyer, Andreas Pokorny, Geoffrey Romer, Ryan Shepherd, Bjarne Stroustrup, Tony Tye, Tony Van Eerd, Ville Voutilainen, Titus Winters, and Michael Wong.

2 Why do something: Problem description, and root causes

2.1 Exceptions have not replaced error codes, and vice versa

“There are still people who argue against all use of exceptions and people who claim that exceptions should be used consistently instead of error codes.” — [P0939R0]

Exceptions are the error handling model that is required by key parts of the language (for constructors and operators) and by the standard library, but are widely banned. This means that a large fraction of the C++ community is not actually using ‘real’ C++, but are using a language dialect, and either a nonstandard library or none at all.

Even though exceptions are required, and have been available for some 25 years, they have not replaced error codes for error handling in C++. Therefore, they never will unless they are changed in some way to address the reasons they cannot be used universally (see §2.5, “Root causes”). The community are voting with their feet:

- Major coding guidelines ban exceptions, including common modern guidelines endorsed by the world’s top advocates of C++ exceptions. For example, the Google C++ Style Guide [GSG] bans exceptions. The Joint Strike Fighter Air Vehicle C++ Coding Standards (JSF++) [JSF++ 2005] was produced by a group that included Bjarne Stroustrup and is published on Stroustrup’s personal website, and bans exceptions.
- Many projects ban exceptions. In [SC++F 2018], 52% of C++ developers reported that exceptions were banned in part or all of their project code — i.e., **most** are not allowed to freely use C++’s primary recommended error handling model that is required to use the C++ standard language and library.
- The C++ Core Guidelines’ Guidelines Support Library [GSL] requires exceptions, and cannot be used in such projects. We are already getting requests for a nonthrowing version of GSL, which changes some of its interfaces (e.g., `narrow` reports errors by throwing `narrowing_error` and would have to change).
- Non-throwing dialects of the STL and the rest of the standard library proliferate, and C++ implementation vendors continue to receive requests to support those nonstandard dialects.
- Every C++ compiler supports a mode that disables exception handling (e.g., `-fno-exceptions`).

This is an intolerable rift: Large numbers of “C++” projects are not actually using standard C++.

But switching to error codes isn’t the answer either — error codes cannot be used in constructors and operators, are ignored by default, and make it difficult to separate error handling from normal control flow.

2.2 Instead, we’re actively proliferating dual interfaces that do *both*

“Filesystem library functions often provide two overloads, one that throws an exception to report file system errors, and another that sets an `error_code`.” — [N3239]

Because we cannot universally recommend either exceptions or error codes, the community and even the committee are proliferating dual error reporting interfaces that support both, by providing throwing and non-throwing alternatives. Worse, the ‘non-throwing’ alternatives in the standard are only non-throwing for some kinds of errors, and still also throw to report other errors from the same function.

For example, the C++17 `std::filesystem` library supports reporting *file system errors (only)* as either exceptions or as error codes, often providing a pair of functions, one for each style; both functions still report non-file errors using exceptions. For example, consider `std::filesystem::directory_iterator::operator++`:

```
directory_iterator& operator++();
directory_iterator& increment( std::error_code& ec );    // note: NOT noexcept
```

Note `noexcept` was removed from the second function, and a number of similar ones, at the recent Jacksonville meeting. See [LWG 3013] and [LWG 3014]. Before that it was already absent for over a dozen similar functions per the policy summarized in the next paragraph.

The current design policy for `filesystem` is that, for file system codes only (which can also be just status codes), the first function of each such pair reports them using exceptions and the second reports them using `error_code`; both alternative functions can still throw exceptions for other non-file errors. This means that inside this dual error reporting design (two alternative functions) is a second nested dual error reporting design (in the same function, some errors are reported via exceptions and others via `error_codes`), and this is intentional.

Notes This has surprised a lot of people, including SG14 in [P0824R1] sections 4.2 and 4.9. I find that programmers who encounter the `filesystem` API make the assumption that the second alternative is for programs that don't want to throw exceptions or that don't have exception handling turned on. So it is important to teach all `filesystem` users that `filesystem` does not actually generally support a non-throwing mode, despite the overloads that appear to do so. — Rather, the motivation to not throw exceptions appears to be more because “routine” status is reported using the same code type that also communicates true errors (see §1.1 point (1) and accompanying Note, in this paper), and so callers that use exceptions exclusively were found to be often littered with local `try/catch` blocks to handle “routine” events. A different design approach for libraries like `filesystem` would be to distinguish “routine”/informational status codes as distinct (a separate object) from error codes, and report only the latter as errors.

We are following this dual design policy even though we know it has serious drawbacks:

- **(worst) Makes error handling harder.** It's hard enough to get call-site programmers to perform consistent and correct error handling when given a single consistent method of error reporting. Now we're giving them two alternatives to choose from — and then in one of those alternatives additionally reporting errors in two ways from the same function, asking callers to write two error handling twice using different styles. This makes it difficult for callers to write reliable code.
- **Interface bloat.** It bloats the library's interface, which includes both the library's documentation (external) and an implementation's test burden (internal).
- **Encourages dialects (I).** It actively encourages C++ dialects, because some callers use exceptions and some use error codes, and both are actively supported by the API.
- **Inconsistency.** It eliminates the ability to use a consistent function name at least for operators since these cannot be overloaded in this way (e.g., `operator++` vs. `increment`, above).
- **Encourages dialects (II): Discourages other language features.** It creates a broader ripple effect through the language by adding a reason to avoid otherwise-recommended unrelated features (e.g., C++ overloaded operators).

Despite all these drawbacks, within the C++ committee we are now having active discussions, not about solving the underlying problem so we can stop creating dual interfaces, but instead about applying this pattern to still more parts of the standard library (e.g., networking). The above example of `directory_iterator::operator++` also acknowledges implicitly that even the standards committee agrees that exceptions are not considered sufficient even in the cases where they have the strongest possible language advantage over error codes, namely for operators and constructors. No other language I know of has such a bifurcation problem.

2.3 Worse, for some real-world code *neither* is acceptable

“Recent threads on the Boost email-list continue to highlight the ongoing confusion and disagreement even over the proper or idiomatic use of `std::error_code` ... One might think such discussions should by now be resolved; but no...” — [Bay 2018]

“On table-based exception handling implementations... A throw...catch cycle is always at least thousands of times more expensive than a return statement, and always must be so, even when the throw...catch is inlined” — [Douglas 2018]

In some real-world code, neither an exception nor an error code is acceptable. A poster child example is a constructor or overloaded operator that can fail, and must be usable in memory-constrained and/or real-time code:

- **It cannot use exceptions**, because the space and time cost of today’s dynamic exception handling is nondeterministic and so cannot be guaranteed to fit in bounded space or bounded time. This is why exceptions are banned in JSF++ [JSF++ 2005] and the Mars Rover flight software [Maimone 2014].
- **It cannot use error codes**. For constructors, using error codes means embracing a poor and incompatible C++ dialect, either pervasively using two-phase construction and “is-constructed” tests on every type with a fallible constructor (for example, see [Outcome 2.0: Result returning constructors](#)) or replacing constructors with factory functions. For operators, using error codes means not using operators at all but replacing them with named functions (for example, see the preceding `std::filesystem` example which renames `operator++` to `increment`).

Yet the standard library itself, including STL, specifies constructors and operators that can fail. So we cannot easily use a conforming standard library in memory-constrained and/or real-time code; that would require modifying it to report errors in another way (and in an incompatible dialect of C++, per above), or leaving its design as-is but applying the hammer of disabling exceptions and just ignoring errors (unacceptable in general).

2.4 And we’re proliferating new patches and alternatives

“Note that expected can also act as a bridge between an exception-oriented code and a nothrow world.” — [P0323R3]

“Exception throwing is absolutely at the heart of Outcome. That’s why Outcome != Expected” — N. Douglas, quoted in [Bay 2018]

Proliferation of patches to make error codes better. We have ongoing active discussions, such as in SG14, about “exception-less error handling” using C++11 `std::error_code` or an evolution thereof (see [P0824R1]). Also, C++17 added the `nodiscard` attribute for “important” return values, motivated in part by returned status information that should not be ignored (see [P0068R0] example 4). Note that `nodiscard` is broadly useful and desirable; however, the specific use of relying on it to make sure callers don’t silently ignore errors is a “patch” in terms of the error handling model.

Proliferation of new library-only solution attempts. The C++ committee and community continue to consider new alternatives in new standardization. For example:

- In the committee, we are advancing the proposed `std::experimental::expected<SuccessResult, ErrorResult>` [P0323R3]. As noted in the paper: “C++ already supports exceptions and error codes, *expected* would be a third kind of error handling.”
- Boost, while aware of this, continues to pursue evolving a distinct `outcome::result<SuccessResult>` with different tradeoffs, notably lower run-time overhead than expected for *expected*. From the 2018 Boost acceptance report for Outcome v2 [Bay 2018], emphasis original: “The prime motivation for acceptance is: Reviewers have real-world use cases **today** for which they found Outcome to be an effective and best available alternative; and which is consistent with current-need and expectations; and which is consistent with ongoing C++ Standard evolution efforts. From the Library Author: ‘Outcome is really an abstraction layer for setting per-namespace rules for when to throw exceptions. Exception throwing is absolutely at the heart of Outcome. That’s why Outcome != Expected, and why it ICEs older compilers, and why C++ 14 is needed.’ ”

As library solutions without language support, these approaches have two major problems: First, they are fundamentally attempts to regain use of the return value for error reporting, and by fusing “success” and “error” returns they force callers to perform error handling using only normal control flow constructs to inspect a merged value. Second, they contribute to fracturing C++ error handling because they are adding a third or a fourth style; for example, in their current form, it is not clear whether these would be universally adoptable throughout `std::filesystem` to resolve its dual-mode problem, and [Bay 2018] includes the note that Outcome is not intended to be usable for all error handling.

The good news is that these efforts are blazing trails, and converging, in a good direction: They are already very close to expressing a library type that is suitable for universal C++ error reporting, with strong efficiency and fidelity characteristics. That’s important, because it means we may now be at a point where the library type is sufficiently baked for the language to be able to embrace it and help them (this proposal).

Note There are two families of use cases given for *expected*, and only one is about error handling: (1) `expected<T1, T2>` where both paths are normal “routine” control flow, and `T2` is an alternate result for a “routine” outcome; for this, the authors of *expected* acknowledge that `variant<T1, T2>` might be a more natural choice. (2) `expected<T, E>` where `E` really represents an error; for this, I think there is real benefit in this paper’s proposed language support to keep the error-handling paths distinct and automatically propagate the errors.

2.5 Root causes: Why we can’t just use exceptions everywhere today

“I can’t recommend exceptions for hard real time; doing so is a research problem, which I expect to be solved within the decade” — [Stroustrup 2004]

Above, we enumerated the performance issues with today’s dynamic exception handling model: binary image bloat, run-time cost, and deterministic run-time space and time cost (when throwing).

The root cause of these problems is that today’s dynamic exception handling model violates two of C++’s core principles, zero-overhead and determinism, because it requires:

- throwing objects of *dynamic types*, which requires dynamic allocation and handling; and
- using *non-local by-reference* propagation and handling semantics, which requires non-local coordination and overheads, and requires arbitrarily many exceptions with distinct addresses at the same time.

For additional details beyond what is covered below, see section 5.4 of the Performance TR, [ISO 18015:2004].

(1) Today’s exception handling is not zero-overhead (binary image size, run-time space and time). Exception handling is one of two C++ language features that violates the zero-overhead principle, that “you don’t pay for what you don’t use” and that “when you do use it you can’t reasonably write it more efficiently by hand.” For example, just turning on exception handling support in a project previously compiled without exception support — i.e., one that is not yet throwing *any* exceptions at all — commonly incurs significant binary space overhead; I regularly hear +15% reported (Chris Guzak in personal communication regarding Windows internal examples, and +16% [reported by Ben Craig](#) on the SG14 mailing list for a different code base and environment), and I have recently seen other Windows internal examples with +38% bloat, down from +52% after recent additional back-end optimization (Ryan Shepherd, personal communication). The overhead arises in various places: In the binary image, we have to store jump tables or other data/logic. At run time, most implementations reserve additional stack space per thread (e.g., a 1K reservation, to save a dynamic allocation) and require and use more-expensive thread-local storage.

(2) Today’s dynamic exception handling is not deterministic (run-time space and time determinism). This is the primary reason exceptions are banned in many real-time and/or safety-critical environments (for example, many games, coding standards like JSF++ [[JSF++ 2005](#)], and environments like the Mars Rover flight software [[Maimone 2014](#)]). C++ allows there to be multiple active exception objects of arbitrary types, which must have unique addresses and cannot be folded; and it requires using RTTI to match handlers at run time, which has statically unpredictable cost on all major implementations and can depend on what else is linked into the whole program.¹ Therefore during stack unwinding the exception handling space and time cost is not predictable as it is with error codes. Adequate tools do not exist to statically calculate upper bounds on the actual costs of throwing an exception.

2.5.1 Examples of inherent overheads

Here are some specific examples of required overheads.

Note that all of the overhead examples in this subsection are inherent in the model of “throwing dynamic types using non-local by-reference propagation” — the costs cannot in general be avoided simply by using a smarter implementation strategy (they can only be moved around, such as by using table-based vs. frame-based implementations, or by using heap vs. pin-the-dead-stack allocation), and they cannot in general be optimized away (even with heroic potential optimization efforts that implementations do not actually attempt today).

Note There have been extended arguments about whether the choice of table-based vs. frame-based exception handling implementation strategies might be the reason why exceptions have not been universally adoptable. It isn’t. For details, see section 5.4 of the Performance TR, [[ISO 18015:2004](#)]. — Briefly: Table-based implementations are better when failure almost never happens, and frame-based shines when failure is common, but both still incur non-local costs just to enable exceptions regardless of whether, or how often, they are thrown, and both incur some cost even on the success path. Neither implementation style can achieve zero-overhead or determinism, because the costs are inherent in exceptions’ demands for additional binary image code/data, run-time heap allocation, and dynamic run-time typing — table-based vs. frame-based is just moving those costs around, not eliminating them.

(1) Today’s dynamic exceptions can require arbitrarily many exceptions in flight with unique addresses. Handling an exception can cause additional exceptions (of potentially unrelated dynamic types) to be thrown from the `catch` handler before the exception being handled can be destroyed. Multiple exceptions in flight cannot be

¹ Absent heroic optimizations, such as fully inlining all functions called from a `catch` block to prove there is no re-throw.

folded using normal optimizations for variable folding, and so because arbitrarily many exceptions can be in flight, and their number is not in general statically predictable, throwing an exception requires arbitrary amounts of memory.

(2) Today’s dynamic exception objects cannot be allocated normally in the local stack frame. This leads to unpredictable time space and/or time costs in various ways. Here are two typical examples:

- On platforms that use the [Itanium ABI](#), exceptions are required to be allocated on the heap (modulo potential optimizations that are not actually implemented today, such as the proposed LLVM optimization in [\[Glisse 2013\]](#)). Heap allocation requires unpredictable time, even on allocators that avoid global synchronization in the memory allocator.
- On Windows platforms, exceptions are technically allocated on the stack, but they are far from normal stack allocations: When an exception is thrown, the stack contents are destroyed by unwinding back to the `catch` handler, but the now-unused stack space itself is not yet deallocated until the handler ends — in effect the stack storage is “pinned” until the original exception can be destroyed. This means that the `catch` handler code must run at a stack location *beyond* the stack depth where the being-handled exception was thrown, skipping the dead space — and this repeats recursively for any additional exceptions thrown during handling, and C++ today allows arbitrarily many such exceptions to be created (see previous point). For a simple example of just three such in-flight exceptions and how they multiply stack usage, see the Appendix. In this implementation strategy, the stack memory usage is therefore a total of the individual stack depths of each path that threw an exception while another exception was already active, and I do not know of tools that compute a static memory use bound. (I have not tried to measure whether this attempt at ‘in-stack-memory-but-not-really-stacklike’ allocation is typically better or worse overall than just doing a real heap allocation; it will nearly always be worse in total memory consumed, but it does avoid contention on the global allocator.)

(3) Therefore, today’s dynamic exceptions cannot share the return channel. When an exception is thrown, the normal return channel is unused — it is entirely wasted. That itself is an inherent architectural pessimization.

(4) Today’s dynamic exceptions require using some form of RTTI to match handlers. The cost of RTTI is generally nondeterministic in both space and time.

Note RTTI is the other C++ language feature that violates the zero-overhead principle. We need to fix RTTI too, but this is not that paper. However, because exceptions rely on RTTI (by propagating and manipulating dynamically typed exceptions) so that the cost of RTTI is indirectly part of the exception handling cost, here is a brief summary of why RTTI violates zero-overhead and the two issues that most directly affect exception handling: ²

First, it requires support for `typeid`, including `typeid.name()`, which is effectively metadata. Normally C++’s zero-overhead design rejects “pay for what you don’t use” overheads that add space or time cost even when not used; the usual poster child examples are “always-on” or “default-on” (a) metadata (e.g., we have always rejected storing even the names of enumerators) and (b) garbage collection (e.g., we support it via opt-in libraries but not as the global default). The one place C++ adds required metadata is in `typeid`, especially `typeid.name()`.

² There are other issues less directly relevant to exception handling. For example, in addition to these overheads, some implementations of `dynamic_cast` incur needless extra run-time inefficiencies, such as by performing textual string comparison as part of the cast operation. Those overheads can be fixed to incrementally improve RTTI performance, but those fixes are not germane here because they don’t materially change the RTTI impact on exception handling.

Second, it does not distinguish dynamic casts that have different costs. For example, the following have different power and different costs (and are already distinguished in the [Itanium ABI](#)): (a) downcast to the statically unknown most-derived type (complete object); (b) downcast to a statically known derived type (not necessarily most derived); (c) cross-cast to a statically known sibling type; and (d) upcast from the statically unknown most-derived type to a public statically known base type. Because `dynamic_cast` must perform all of the first three operations (a) (b) and (c), it is necessarily at least as expensive as the most expensive of all three. (See [\[O'Dwyer 2017\]](#) for a lucid treatment.) Exception handling only needs (d).

To fix the above two issues (in the context of enabling a more efficient dynamic exception handling implementation), we could provide a version of RTTI for `catch` implementation use only that is not disabled when RTTI is otherwise disabled, and that does not include `typeid` support and includes support only for dynamic casting of type (d), with the caveat that (d) might still violate either the zero-overhead principle (either by generating additional static data in the vtable to enable constant-time casting as demonstrated in [\[O'Dwyer 2017\]](#) slides 40-42, or by avoiding additional static data at the cost of non-constant-time casting which would leave it unsuitable for real-time code).

Third (and this might or might not be able to be mitigated by the approach in the previous paragraph), the cost of RTTI can be effectively unpredictable because linking in unknown third-party shared libraries can dramatically affect the performance of RTTI lookup, and thus the performance of exception handling. In general we cannot predict whether some end user, or even customer of that end user, will not combine our code with some other code in the same process; Niall Douglas reports real-world cases where a user's linking in other code caused the cost of `throw...catch` to rise dramatically (e.g., 500ms on a heavily loaded machine) due to the environment-specific unpredictability of the RTTI cost.

For the above reasons, major projects and guides (e.g., Firefox, Chrome, the Google C++ Style Guide [\[GSG\]](#)) actively discourage or ban using RTTI and `dynamic_cast`. This usually means that these projects cannot use exceptions either, because today exceptions rely on RTTI.

The projects work around their lack of `dynamic_cast` by using `static_cast` downcasts, using a visitor pattern, or rolling their own homegrown dynamic casting method (e.g., storing a type tag for a known class hierarchy, which does not scale universally). This continues to cause new C++ code security exploits due to type confusion vulnerabilities, where the root cause analysis of many recent security incidents has observed that the code should have used `dynamic_cast`, but did not because of its binary image space and/or run-time costs (for example, see [\[Lee 2015\]](#), paragraphs 2 and 3).

It is an open research question whether C++'s currently specified RTTI is implementable in a way that guarantees deterministic space and time cost. [\[Gibbs 2005\]](#) describes an approach to get constant-time dynamic casting in constrained class hierarchies by having the linker assign type identifiers, but it does not support dynamic libraries or hierarchies or arbitrary shape and size, and so is not a general solution. The two [known](#) followup papers [\[Dechev 2008\]](#) and [\[Dechev 2008a\]](#) did not attempt to address those issues, but focused on contributing improvements in the heuristic for generating type identifiers.

See also:

- §4.6.1: "Wouldn't it be better to try to make today's dynamic exception handling more efficient, *instead of pursuing a different model?*"

- §4.6.2: “But isn’t it true that (a) dynamic exceptions are optimizable, and (b) there are known optimizations that just aren’t being implemented?”

Fortunately, having exception handling with automatic propagation does not require a model with these properties. We have existing counterexamples: For example, although today’s C++ dynamic exception handling is not isomorphic to error codes, Midori’s [\[Duffy 2016\]](#) and CLU’s [\[Liskov 1979\]](#) exception handling models are isomorphic to error codes which enables more efficient implementations, and does not preclude automatic propagation.

3 What we want: Ideal error handling characteristics

3.1 Summary of the ideal: We need exceptions' programming model

“Conflating error handling and control flow is a crime against readability and conciseness.”

— Michael Novak, personal communication

This section lays out what I believe are ideal error handling characteristics. They are not unique to C++; I believe they apply to most modern languages.

Ideal	Exceptions	Error codes	expected<T,E>	outcome<T>
A. “Error” flow is distinct from “success”				
When raising (distinct from normal <code>return</code>)	Yes (<code>throw</code>)	No	Partial (<code>return unexpected</code>)	Yes (return success vs. return failure)
When handling (distinct from success code)	Yes (<code>catch</code>)	No	Partial (<code>.value()</code> throws)	Partial (policy determined)
B. Error propagation and handling				
Errors can only be ignored explicitly (not ignored silently by default)	Yes	Partial (<code>nodiscard</code> , <code>warnings</code>)	No (in current proposal)	Partial (policy configurable)
Unhandled error propagation is automated	Yes	No	No	No
Unhandled error propagation is visible	No (Yes if §4.5)	Yes	Yes	Yes
Writing an error-preserving error-neutral function is simple	Yes	No	?	Yes
C. Zero-overhead and determinism				
Stack allocated (no heap)	No (Yes if §4.1)	Yes	Yes	Yes
Statically typed (no RTTI)	No (Yes if §4.1)	Yes	Yes	Yes
Space/time cost equal to <code>return</code>	No (Yes if §4.1)	Yes	Yes	Yes
Space/time cost fully deterministic	No (Yes if §4.1)	Yes	Yes	Yes

Note This paper does not address other potential improvements that would require a source breaking change, such as that function declarations should default to “does not fail.” In the future, I hope to bring proposals to address those issues in the broader context of exploring how to take a source breaking change that could change defaults and in other ways enable further C++ simplification, but they are beyond the scope of this paper.

Group A: “Normal” vs. “error” is a fundamental semantic distinction, and probably the most important distinction in any programming language even though this is commonly underappreciated. Therefore, the distinction should be surfaced explicitly (though as elegantly as possible) in language syntax and program structure.

Group B: True errors (as opposed to partial-success or other success-with-info) are important and should be handled even if by explicitly doing nothing. Any approach that allows them to be silently ignored will incur long-term cost to program robustness and security, and to a language’s reputation. Further, they should be propagated in a way that the programmer can reason about. — The one place that exception handling fails the ideals shown here is that exception propagation between the `throw` site and the `catch` handler is invisible in source

code, which makes exception-neutral code (which predominates) harder to reason about and is primarily addressed by widespread use of RAII stack-based variables (which are good for many reasons besides exception safety).

Group C: This group is “because this is C++,” but it’s also where exception handling most falls short today. The proposal in §4.1 is motivated by the observation that the costs are associated with being able to throw arbitrarily typed exceptions.

3.2 Goals and non-goals

This paper aims at two essential goals, that we must achieve to keep C++ unified (whether via this proposal or in some other way).

(1) We must remove all technical reasons for a C++ project to disable exception handling (e.g., by compiler switch) or ban use of exceptions, in all or part of their project. This does not mean requiring a project to actually use exceptions for all their error reporting. It just means that every C++ project be able to use the standard C++ language and a conforming standard library.

SG Poll The 2018-04-11 SG14 telecon took a poll on whether the above is a problem worth trying to solve: **Unanimous consent.**

(2) We must reduce divergence among error reporting styles. This means converging as many of the divergent error reporting styles as possible by providing a usable model that can subsume some of the others.

Non-goals (but we might effectively reach them anyway, at least in part):

- It is not a goal to make exceptions safe for propagation through C code. — However, because this proposal defines a kind of exception that is implemented as an error return, I believe this proposal could make it possible for C and other-language code to correctly invoke C++ functions that use the proposed exception model to report errors and that otherwise are written in the C subset.
- It is not a goal to enable errors to be handled using normal control flow constructs. — However, §4.5 describes how this proposal puts us on a path where programmers can write code in exactly the same style as using `expected<T,U>` today, but keeps the normal and error paths as fully distinct (`catch` instead of using normal control constructs).
- It is not a goal to enable distantly-handled errors to contain arbitrary *programmatically*-usable information. Distantly-handled error details primarily need to be *human*-usable (e.g., debugging and trace logging), and a `.what()` string is sufficient.

4 How to get there: Throwing values

... The [Outcome] Library Author can be congratulated (or scolded) for exploring work or attempting Boost community review in such a contentious space.” — [Bay 2018]

This section proposes a solution — not without trepidation, because I understand this touches an electrified rail.

Error handling is perhaps the most fundamental design point of any programming language. It cannot be changed lightly. However, if there are chronic unresolved issues with error handling, then addressing those successfully can have outsized leverage to deliver broad improvement across all uses of the language — if we can design for backward source and binary compatibility, so that new and old code can interoperate seamlessly.

4.1 Core proposal: `throws` (addresses §3.1 groups C and D)

SG Poll The 2018-04-11 SG14 telecon took a poll on pursuing this direction: 12-2-0 (Favor-Neutral-Against).

4.1.1 Elevator pitch

This proposal aims to marry the best of exceptions and error codes: to allow a function to declare that it *throws values of a statically known type*, which can then be implemented exactly as efficiently as a return value.

Throwing such values behaves as-if the function returned `union{R;E;}+bool` where on success the function returns the normal return value `R` and on error the function returns the error value type `E`, both in the same return channel including using the same registers. The discriminant can use an unused CPU flag or a register.

The entire implementation of throwing and propagating such exceptions is entirely local within a function and its stack frame (no need for separate tables, no separate allocation outside the stack frame), is statically typed (no need for RTTI), and is equally deterministic in space and time as returning an error code. It is at least zero overhead compared to returning an error code, and can be negative overhead in practice compared to returning an error via an `error_code&` out-parameter because an out-parameter cannot share the return channel.

Expanding the elevator pitch to specific audiences:

- **If you love exceptions, including you wish you could use exceptions but can’t tolerate their cost:** This is exception handling, with error handling separated from normal control flow and automatic propagation and never-silently-ignorable errors — plus the special sauce that if you agree to throw an `error` value you get a more efficient implementation that is truly zero-overhead and fully deterministic in space and time.
- **If you love `expected/outcome`:** This is embracing `expected/outcome` and baking them into the language, the function always returns exactly one of `R` or `E` — plus the special sauce that you get automatic propagation so you don’t have to manually return-up the results, and with a distinct language-supported error path so that callees can write `throws` (instead of `return unexpected`) and callers get to cleanly put all their error handling code in distinct `catch` blocks (instead of `if(!e)` blocks) while still writing in the same basic `expected` style (see §4.5).
- **If you love error codes:** This is just giving a function two return paths, one for success and one for failure where the latter returns an error code as usual — plus the special sauce that the language lets you distinguish the two, the error code doesn’t monopolize your natural return value channel, you don’t have to propagate the error by hand, and you can’t forget to check errors.
- **If your project needs fail-fast on all heap exhaustion:** See §4.3.3 and §4.4.

4.1.2 `std::error` type

“By allowing multi-level propagation of exceptions, C++ loses one aspect of static checking. One cannot simply look at a function to determine which exceptions it may throw.” — [Stroustrup 1994] p. 395

Let *relocatable* mean movable with the semantics that the destructor of the moved-from object is never called.

Let *trivially relocatable* mean that the move step is trivial (but the destructor need not be trivial).

Notes “Trivially relocatable” implies that, given two objects `src` and `dst` of type `T`, performing a move from `src` to `dst` followed by performing destruction `src.~T()` is functionally equivalent to just copying the bytes from the source object to the destination object. A roughly-equivalent formulation is that moving `src` to `dst` is functionally equivalent to just copying the bytes from `src` to `dst` and then copying the bytes from a default-constructed `T{}` to `src`.

Any trivially copyable type is also trivially relocatable, but many types are trivially relocatable without being trivially copyable, including (in most implementations) `unique_ptr`, `exception_ptr`, and `string`.

See also the directly related `[[move_relocates]]` proposal [P1029R0]. If that proposal is adopted, `std::error` can be annotated using the general `[[move_relocates]]` mechanism. In the meantime, for this paper I define the term only in order to define `error` itself as a type having trivially relocatable semantics, and to define the destructor treatment of a user-selectable error type `E` in §4.6.5 if it is/isn’t relocatable.

There have been suggestions for such a general language feature, under names such as “destructive move,” but neither this proposal nor [P1029R0] proposes that.

[O’Dwyer 2018a] demonstrates `is_trivially_relocatable` as an opt-in library tag, where making the libc++ implementation of `vector<unique_ptr<int>>` relocation-aware, and tagging `unique_ptr` as relocatable, improved `.reserve()` reallocation performance by 3×.

See [this Godbolt example provided by Niall Douglas](#) which demonstrates that having either a trivial move constructor or a trivial destructor is sufficient to return `error` in registers on the Itanium ABI. Using the related new (2018-02) Clang extension `[[clang::trivial_abi]]` (see [O’Dwyer 2018b]) it is possible to get register-passing capability for a wider variety of RAII types; see [this Godbolt example provided by Arthur O’Dwyer](#).

The single concrete type `error` is an evolution of `std::error_code`; see also related paper [P1028R0]. It has the following ideal requirements, including the improvements suggested by SG14’s review in [P0824R1]:³

- It always represents a failure (there is no `0` success value). A default constructor would construct a general or “other” nonspecific error value.
- Its size is no greater than two pointers, typically a “payload” (usually an integer) plus a `constexpr` “domain” (usually a pointer or hash value that is used only for its type to distinguish the domain).

³ See [Douglas 2018c] for a sample prototype implementation, which claims to meet all of the requirements stated in this list. It is a refinement of `system_code` (an alias for `status_code<erased<intptr_t>>`) from [Douglas 2018a], which itself is just starting to be brought to Boost and SG14.

- Its “domain” discriminant (similar to `std::error_category` but with the improvements suggested in [P0824R1]) is able to represent all causes of failure in the C++ standard library, as well as POSIX system codes, Windows `NTSTATUS`s, COM `HRESULTS`s, and other popular error reporting systems.
- It is type-erased, allocation-free, trivially relocatable, constant-time in all operations, ABI-safe, and safe to use in header-only libraries, while also non-lossy to preserve the original cause of failure.
- It provides `weak_equality` heterogeneous comparison that performs *semantic equivalence* comparison across domains, which aids composability; for example, “host unreachable” errors from different domains (e.g., Win32 and POSIX) compare equivalent to each other and to `errc::host_unreachable` which can be queried in portable code without being dependent on the platform-specific source error.

4.1.3 throws static-exception-specification

This paper proposes that a function (including lambda function) may declare a **static-exception-specification** of just `throws` to indicate that the function can fail. If the function fails, it throws an object of type `std::error` implemented as-if returning it as an alternate return value (i.e., on the stack).

For example:

```
string f() throws {
    if (flip_a_coin()) throw arithmetic_error::something;
    return "xyzzys" + "plover";           // any dynamic exception is translated to error
}

string g() throws { return f() + "plugh"; } // any dynamic exception is translated to error

int main() {
    try {
        auto result = g();
        cout << "success, result is: " << result;
    }
    catch(error err) {                     // catch by value is fine
        cout << "failed, error is: " << err.error();
    }
}
```

Note I considered using `throw`, but we also want a “does this throw static exceptions” operator analogous to the `noexcept` operator (see §4.1.4), and we can’t use `throw` unambiguously for that operator. So for consistency between this declaration and the operator, I am using the strawman syntax `throws`. Using `throws` also helps to avoid any confusion with the mostly-removed dynamic exception specification `throw(...)` syntax.

For a function `f` declared with a static-exception-specification `throws`:

- All declarations of `f` must be declared `throws`, including in base classes if `f` is a virtual override.
- `f` behaves as-if `noexcept(false)` when queried by the `noexcept` operator and the `*noexcept*` and `*nothrow*` traits (e.g., `move_if_noexcept`, `is_nothrow_move_constructible`).
- Conceptually, in the case of failure `f` behaves as-if it were declared with a return type of `error`. The normal and error returns share the same data channel and exactly one is used.

Notes This includes being able to return `error` in registers. There are no functions declared with `throws` today, so we have an opportunity to define the ABI for this new set of functions, as a new case that extends existing calling conventions. For example, we can expand the number of registers (e.g., to 4 or 6 on x64, to 8 on AArch64), and use one of the unused CPU flag bits to indicate whether those registers contain a value or an error.

An alternative would be to formally specify this be implemented as an `E*` “out” parameter, so that if the function is otherwise callable from C (or other languages that understand C as *de facto lingua franca*) then the error handling is consistently available from calling code in those languages. An out-parameter implementation strategy could generate more compact code for exception-neutral code, and reduce total stack usage. — We will prototype and measure both alternative implementations.

- For any throw-expression in `f`’s body that has no argument (i.e., `re-throw`); It must appear in a `catch` block and behaves as-if `throw e`; where `e` is the catch block parameter.

Note Alternatively, for an `error` value only, we could disallow anonymous `re-throw` and require `throw e`; . But then we would still want to support anonymous `re-throw` as a synonym in migration/compatibility mode (see §4.7).

- For any throw-expression in `f`’s body that has argument, `throw expr`; , where `expr` is convertible to `error`:
 - If `f` has a local `catch(error)` or `catch(...)` handler between the call site and the end of the function, control goes to that handler as-if via a forward `goto`.
 - Otherwise, it behaves as-if `return expr`;

Notes The usual rules apply, such as that if `expr`’s or `e`’s type is not `error` or convertible to `error` then throw-expression is ill-formed.

This specification is deliberately in terms of forward-`goto` semantics (zero overhead by construction), not in terms of a notional try-catch where we then rely on optimizers to elide the overhead (attempting to claw back zero overhead by optimization).

- When calling another function `f2` that also has a static-exception-specification and that throws an exception `e`, the effect is as-if `throw e`;
- When one of today’s dynamic exceptions is unhandled in `f`’s body, regardless of whether it originated from a nested function call or a `throw` statement throwing a dynamic exception, the exception is automatically caught and propagated: If the caught exception is of type `error`, we just return it. Otherwise, it is translated to an `error` with a meaningful value for all `std::` exception types; for example, `bad_alloc` would be translated to `std::errc::ENOMEM`. Otherwise, we can additionally store as payload a raw pointer to an `exception_ptr` to the dynamic exception (see §4.6.4), without sacrificing trivial movability.

Notes The mapping from exception types and values to `error` values will need to be fully specified in the standard.

We should include a customization point to allow enabling automatic translation also for other exception types.

- If `f` is a virtual function, then: Every base function that `f` overrides must be declared `throws`. Every further-derived override of `f` must be declared either `throws` or `noexcept`.

Note If a base function is declared `noexcept`, a derived override must also be declared `noexcept`.

For a function declared without a static-exception-specification:

- When calling another function `f` that has a static-exception-specification and that throws an exception `e`: If the `error` is a wrapped `exception_ptr`, it rethrows the dynamic exception. Otherwise, if the `error` value corresponds to one of the meaningful values for a `std::` exception type, it throws an exception of that type; for example, `std::errc::ENOMEM` would be translated to `bad_alloc`. Otherwise, the effect is as-if `throw e;`, that is, it throws the `error` itself as a dynamic exception.

Notes Today, implementers are permitted, but not required, to make `exception_ptr` trivially relocatable. If it is, `error` can hold an `exception_ptr` directly as its payload. If it is not, the `exception_ptr` can be allocated on the heap and `error` can hold a raw pointer to it and destroy it when it's done with it (at the end of the `catch` handler that consumes it without rethrowing).

Some reviewers have expressed the opinion that it might be better to require code to manually translate between static and dynamic exceptions. The main motivation for automating this is two-fold: (1) We want to make it as easy as possible to upgrade existing non-`noexcept(true)` functions by just adding `throws`. If we don't do this, then the programmer still has to write a `try/catch` by hand. (2) The programmer can't write better code than we could by automatically translating the exception. So, since the `try/catch` is both always necessary and cannot be written more efficiently by hand, it should be default and automatic. — That said, tools could warn when such implicit translation is happening, such as to find not-yet-upgraded places in an existing code base.

It may be desirable to allow overloading on `throws` with the meaning that, as a final tie-break, a function with/without a static-exception-specification will prefer another one with/without a static-exception-specification; `throws` would also become part of the function's type as with `noexcept`. This would be primarily useful to optimize the translation of dynamic exceptions to `errors`. For example, in this proposal, a function with a static-exception-specification that invokes today's existing `operator new` and doesn't handle a `bad_alloc` would get it automatically translated to `std::errc::ENOMEM`. A quality implementation that inlines the call to `operator new` could elide the dynamic exception, but that relies on an optimization. If we supported overloading on `throws`, then we could additionally provide a (non-placement) overload of `operator new` that is decorated with `throws`, and it will be used by `throws` functions to guarantee no dynamic exception ever happens (whether the call is inlined or not) while leaving all existing uses unchanged (all existing code still uses the existing `operator new`).

4.1.4 Conditional `throws` and `operator throws`

A static-exception-specification of `throws(cond)` has the basic meaning `noexcept(!cond)` but additionally can distinguish between static (default) and dynamic exception reporting. For example:

```
template<class T>
struct X {
    void X(X&&) throws(!is_nothrow_move_constructible_v<T,T>);
    X&& operator=(X&&) throws(!is_nothrow_move_assignable_v<T,T>);
};
```

The condition `cond` evaluates to a value of type `enum except_t { no_except=false/*0*/, static_except=true/*1*/, dynamic_except/*=2*/ }`.

The operator `throws(expr)` performs a compile-time computation returning the `except_t` used by `expr`, and returns `no_except` if the `expr` is declared to be `noexcept`, otherwise `static_except` if `expr` is declared to be `throws`, otherwise `dynamic_except`.

Note We can't use `throw` unambiguously for the operator, hence `throws`. This is the primary motivation for using `throws`, not `throw`, as the keyword for declaring that a function uses static exceptions.

This permits an algorithm such as `std::transform` to efficiently be `noexcept`, report errors using static exceptions, or report errors using dynamic exceptions exactly as the caller-provided operation does:

```
template< class In, class Out, class Op >
Out transform( In first, In last, Out out, Op op ) throws(throws(op(*first)))
```

In this example, each instantiation of `transform` reports error however `op` does, using exactly one of static exceptions, dynamic exceptions, or `noexcept`.

Note A function that wants to adapt to multiple suboperations that could have different error modes (e.g., some could be `no_except`, others `static_except`, and/or still others `dynamic_except`) can compute how it wants to report errors. It is expected that a common preference will be to be `dynamic_except` if any of the suboperations are that, otherwise `static_except` if any of the suboperations are that, else `no_except`. This is one reason why the enumerator values were chosen as shown, so that for such a function the algorithm for combining the suboperation modes is just `std::max: std::max({no_except, static_except, dynamic_except})` does the right thing, and is already `constexpr`.

I expect conditional `throws` to be used less frequently than conditional `noexcept`. Today, conditional `noexcept` has three main uses:

- **(rare, applies also to conditional `throws`) To enable callers to use a different algorithm.** For example, enabling `move_if_noexcept` can allow using a more efficient algorithm while still giving the strong guarantee. The most common use case is for annotating move operations of generic wrappers and containers.
- **(medium, applies also to conditional `throws`) To enable generic intermediate code to preserve the `noexcept`-ness of its implementation.** For example, a `std::transform` call could (but currently is not required to) declare itself `noexcept` if the caller-provided operation is `noexcept`.
- **(very common, does not apply to conditional `throws`) To claw back performance lost to today's dynamic exception handling overheads.** That those overheads are so expensive that we are willing to frequently do tedious programming in the function type system to avoid them (and thereby leak implementation details into the function type) is a strong statement about the unacceptability of today's dynamic exception overheads. In this proposal, new code that otherwise would resort to conditional `noexcept` to avoid the dynamic exception overheads would instead throw statically typed values.

Because conditional `throws` is only for the first two motivations, which are rarer and primarily motivated by generic code that (a) uses move functions or (b) is adaptive to report errors from callees that might use today's dynamic exceptions without converting them to `std::error`, I expect uses of conditional `throws` to be rarer than uses of conditional `noexcept`.

For a function `f` declared with a static-exception-specification `throws(cond)`, then in addition to the rules in §4.1.2:

- `f` must not be virtual.

Note In general, trying to mix the always-static computation `throws(cond)` with the always-dynamic `virtual` appears to be a mismatch. The primary known use case for `throws(cond)` is generic move operations, which should not be virtual.

However, if we do encounter real examples where this is needed, we can specify it by replacing the foregoing bullet with the following:

- If `f` is a virtual function, then: Every base function that `f` overrides must be declared either `throws` or `throws(cond)` with the identical condition. Every further-derived override of `f` must be declared either `throws(cond)` with the same condition or `noexcept`.

4.1.5 Achieving zero-overhead and determinism

“A big appeal to Go using error codes is as a rebellion against the overly complex languages in today’s landscape. We have lost a lot of what makes C so elegant – that you can usually look at any line of code and guess what machine code it translates into.... You’re in a statically typed programming language, and the dynamic nature of exceptions is precisely the reason they suck.” — [Duffy 2016]

Recall from §2.5 that today’s dynamic exception handling model violates the zero-overhead and determinism principles because it requires throwing *objects of dynamic types* and using *non-local by-reference* propagation and handling semantics.

The primary benefits of this proposal accrue from avoiding overheads by design, by throwing *values of statically known types* and uses *local value* propagation and handling semantics, which eliminate the inherent overheads listed in §2.5 because the proposed exceptions are just local return values:

- Multiple exceptions in flight can be folded by routine optimizations (see Appendix for strawman).
- Exceptions are always allocated as an ordinary stack value.
- Exceptions share the return channel instead of wasting it, including being returnable in registers.
- Exceptions have a statically known type, so never need RTTI.

This let us achieve the zero-overhead and determinism objectives:

- Zero-overhead: No extra static overhead in the binary (e.g., no mandatory tables). No dynamic allocation. No need for RTTI.
- Determinism: Identical space and time cost as if returning an error code by hand.

Furthermore, because the proposed mechanisms are much simpler than today’s, they are also more amenable to optimization, including existing optimizations already commonly in use. For example:

- Because exception objects are always entirely local on the stack, and not required to have unique addresses, they are expected to be easy to fold using existing common optimizations.
- In this model, the count maintained by `uncaught_exceptions` is incremented on `throws` and decremented on `catch` as usual, but compensating unread inc/dec pairs are expected to be easier to elide.

Note Whether `uncaught_exceptions` is or can be zero-overhead, including to not use thread local storage, and if not whether to replace it with some other feature that does efficiently support scope guards with zero overhead, is a separable question that does not affect the rest of this proposal.

At call sites (that propagate or handle an error), a potential downside of the if-error-goto-handler implementation model is that it injects branches that can interfere with optimizations. However, because this static exception model is isomorphic to error codes (which the dynamic exception model is not), implementations can also choose whether to implement this exception model as error returns or using table-based handling as a pure optimization (no longer a required overhead). And this was tried out in practice in a large native code system on the Midori project, which used a very similar exception design:

“A nice accident of our model [an the exception model that was isomorphic to error codes] was that we could have compiled it with either return codes or [table-based] exceptions. Thanks to this, we actually did the experiment, to see what the impact was to our system’s size and speed. The exception[-table]s-based system ended up being roughly 7% smaller and 4% faster on some key benchmarks.” — [Duffy 2015]

Again, this is a pure optimization option choice local within a given function’s body. Compilers could provide the option to prefer zero binary size overhead (no tables) or fewer local branches in functions (tables), as with other space/time optimization options.

4.1.6 Side by side examples

To illustrate, consider these examples from [P0323R3] and [Douglas 2018]. In each case, the right-hand side is expected to have identical or better space and time cost compared to the left-hand side, and identical space and time cost as returning an error code by value.

Note In some cases, such as divide-by-zero in the first example, normally it’s best to use a precondition instead. However, I’m preserving the examples’ presented semantics for side-by-side comparison.

Checked integer division, showing a caller that does error propagation and a caller that does error handling. Note that in the bottom row we change the `switch` to testing using `==` which performs semantic comparison correctly if the errors come from different domains.

P0323R3 example	This paper (proposed)
<pre> expected<int, errc> safe_divide(int i, int j) { if (j == 0) return unexpected(arithmetic_errc::divide_by_zero); if (i == INT_MIN && j == -1) return unexpected(arithmetic_errc::integer_divide_overflows); if (i % j != 0) return unexpected(arithmetic_errc::not_integer_division); else return i / j; } </pre>	<pre> int safe_divide(int i, int j) throws { if (j == 0) throw arithmetic_errc::divide_by_zero; if (i == INT_MIN && j == -1) throw arithmetic_errc::integer_divide_overflows; if (i % j != 0) throw arithmetic_errc::not_integer_division; else return i / j; } </pre>
<pre> expected<double, errc> caller(double i, double j, double k) { auto q = safe_divide(j, k); if (q) return i + *q; else return q; } </pre>	<pre> double caller(double i, double j, double k) throws { return i + safe_divide(j, k); } </pre>

```

int caller2(int i, int j) {
    auto e = safe_divide(i, j);
    if (!e) {
        switch (e.error().value()) {
            case arithmetic_errc::divide_by_zero:
                return 0;
            case arithmetic_errc::not_integer_division:
                return i / j; // ignore
            case arithmetic_errc::integer_divide_overflows:
                return INT_MIN;
            // No default: Adding a new enum value causes a compiler
            // warning here, forcing an update of the code.
        }
    }
    return *e;
}

```

```

int caller2(int i, int j) {
    try {
        return safe_divide(i, j);
    } catch(error e) {
        if (e == arithmetic_errc::divide_by_zero)
            return 0;
        if (e == arithmetic_errc::not_integer_division)
            return i / j; // ignore
        if (e == arithmetic_errc::integer_divide_overflows)
            return INT_MIN;
        // Adding a new enum value "can" cause a compiler
        // warning here, forcing an update of the code (see Note).
    }
}

```

Notes Pattern matching would benefit both sides of the last row and remove most of the style delta.

The reason to use `if` and `==` (not `switch`) on the right-hand side is to get semantic comparison. Even so, compilers could warn on missing cases on the right-hand side because all the cases are simple tests against `arithmetic_errc` values, and a simple heuristic can deter whether the code checks for all but one or two of the values in a given domain. In the future, replacing the `if`-cascade with pattern matching would restore the ability for the code to express by construction that it is testing `e`'s values, which would make such diagnostics simpler again as with `switch` today.

[Douglas 2018] example	This paper (proposed)
<pre> outcome::result<int> convert(const std::string& str) noexcept { if (str.empty()) return ConversionErrc::EmptyString; if (!std::all_of(str.begin(), str.end(), ::isdigit)) return ConversionErrc::IllegalChar; if (str.length() > 9) return ConversionErrc::TooLong; return atoi(str.c_str()); } </pre>	<pre> int convert(const std::string& str) throws { if (str.empty()) throw ConversionErrc::EmptyString; if (!std::all_of(str.begin(), str.end(), ::isdigit)) throw ConversionErrc::IllegalChar; if (str.length() > 9) throw ConversionErrc::TooLong; return atoi(str.c_str()); } </pre>
<pre> outcome::result<int> str_multiply(const string& s, int i) { auto result = convert(s); if (result) return result.value() * i; else return result.as_failure(); } </pre>	<pre> int str_multiply(const string& s, int i) throws { auto result = convert(s); return result * i; } </pre>
<pre> outcome::result<int> str_multiply2(const string& s, int i) { OUTCOME_TRY (result, convert(s)); return result * i; } </pre>	<pre> int str_multiply2(const string& s, int i) throws { auto result = convert(s); return result * i; } </pre>

4.1.7 What we teach

What we teach:

- **Function authors:** Prefer to write exactly one of unconditional `noexcept` or `throws` on every function. Dynamic exceptions and conditional `noexcept` still work as well as ever but are discouraged in new/upgraded code.

Note This is similar to how since C++11 the updated guidance for virtual functions is now to write exactly one of `virtual`, `override`, or `final` on every virtual function (e.g., [C.128](#) in the *C++ Core Guidelines*). We got the default “wrong,” but we can now provide a consistent and clear (and mechanically enforceable) style in new code at the cost of writing one word compared to what we could achieve with a time machine.

- **Function callers:** To catch a static exception, write `catch(error)` (note: by value is now fine). The object can be rethrown, copied, stored, etc.
- **Compatibility:** Dynamic exceptions and conditional `noexcept` still work. You can call a function that throws a dynamic exception from one that throws a static exception (and vice versa); each is translated to the other automatically by default or you can do it explicitly if you prefer.

4.1.8 How to migrate, and toolability

To migrate/upgrade, you can adopt `throws` on function declarations incrementally, and the more you use it the more you eliminate overheads:

- To upgrade an existing function that is not `noexcept(true)`, write `throws` on all of its declarations (in place of a conditional `noexcept`-specifier if there was one).
- If your function is on an ABI boundary and currently throws dynamic exceptions, and you have to keep it ABI-stable, then just leave it as-is; you can still start using static-exception-specifications internally within the modules, and they will be translated across the boundary.
- If you have a boundary where you couldn’t use exceptions today because of their cost, and can change the boundary, you can now consider reporting errors using exceptions even on that boundary and still freely call any existing internal functions that still throw the old way.

This proposal is amenable to tool support. To facilitate adoption and migration of a code base:

- Compilers can optionally provide a convenience mode that automatically treats every non-`noexcept(true)` function as though it were declared `throws`.
- “Modernizing” tools can mechanically update every non-`noexcept(true)` function to decorate it with `throws`.

Notes Any transformation to reinterpret or rewrite existing code is an ABI change for that code, so it should be explicitly opted-in to remain under user control.

The standard library can use this proposal in four main stages:

(with no change to the stdlib) Users can use the above transformation to make the standard library available in the new mode.

The standard library could consider specifying that, in freestanding implementations, the standard library is available in the above mode.

Update dual `filesystem` APIs to change from:

```
directory_iterator& operator++();
directory_iterator& increment( std::error_code& ec );
```

to:

```
directory_iterator& operator++() throws;
```

Adopt it in other places, such as new library functions whose efficiency is paramount.

Projects that ban today’s dynamic exceptions for the efficiency and determinism reasons summarized in §2.5 can continue doing so as long as those reasons continue to apply, but now would be able to enable these lightweight exceptions:

- In their own code’s error handling, they can now adopt `throw/try/catch` with all its benefits. Today’s compilers provide modes to turn off today’s dynamic exception handling; they would continue to do that, but provide a mode that enables static-exception-specifications only.
- When using STL and the rest of the standard library, my hope is that this proposal lets them adopt “near-normal” STL with all its benefits even without any change to the standard library specification or implementation. For example, because today the standard library reports error using dynamic exceptions, companies (such as Electronic Arts [N2271] and Windows) have resorted to specifying and maintaining a divergent STL that reports errors in different ways (or fails fast), which changes the interface and design of the library. Instead, with this proposal, the aim is to enable a project to use the compiler mode mentioned above (that automatically treats every non-`noexcept(true)` function as through declared `throws`) and just recompile their existing standard libraries in that mode. The standard library compiled in that mode is not strictly conforming, but it is also nowhere near as divergent as today’s “no-exception STLs” because the delta is much smaller, all errors that were reported by throwing exceptions are still reported by throwing exceptions (just using the `error` type for exceptions), and the transformation from the conforming standard library is purely mechanical.

4.1.9 Discussion: Throwing extensible values vs. arbitrary types

“[In CLU] We rejected this approach [propagating dynamic exceptions] because it did not fit our ideas about modular program construction. We wanted to be able to call a procedure knowing just its specification, not its implementation.” — [Liskov 1992]

“[In Midori,] Exceptions thrown by a function became part of its signature... all ‘checking’ has to happen statically... those performance problems mentioned in the WG21 paper [N3051] were not problems for us.” — [Duffy 2016]

I believe that exceptions of statically known types thrown by value, instead of arbitrary dynamic types thrown by reference, is a natural evolution and an improvement. It follows modern existing practice in C++ (e.g., `std::error_code`, Expected [P0323R3], Outcome [Douglas 2018]) and other languages (e.g., Midori [Duffy 2016]). And, just as when we added move semantics as C++11’s marquee feature, it doubles down on C++’s core strength of efficient value semantics.

We’ve already been learning that it’s problematic to propagate arbitrarily typed exceptions, through experience over the past 25 years not only in C++ but also in other mainstream languages. Consider the following overlapping reasons, in no particular order.

error_code values are more composable than types, easier to translate (when propagating) and compare (when handling). `std::error_code` (especially evolved as described in [P0824R1]) is able to represent and compare equivalent codes from different domains, including all errors reported from the C++ standard library, POSIX, and Windows (with some support for translation between them), plus non-lossy automatic propagation.

Propagating arbitrarily typed exceptions breaks encapsulation by leaking implementation detail types from lower levels. (“Huh? What’s a `bsafe::invalid_key`? All I did was call `PrintText!`”) As a direct result...

The most valuable catches routinely ignore the type information they do have: Distant catches are frequently untyped in practice. It has been observed that the value of exception handling increases with the distance between the `throw` and `catch` (Sergey Zubkov and Bjarne Stroustrup, private communication). But because of the previous bullet, the leaked lower-level types are not useful in practice in higher-level code that does not expect or understand them,⁴ and higher-level code that writes `catch` often resorts to writing `catch(...)` rather than mentioning specific types from lower levels. — Because the most valuable catches (the distant ones) commonly don’t understand the exception’s details and ignore their types, throwing arbitrary types has less value in practice exactly in the cases where automatically propagated exceptions are the more effective.⁵

Propagating arbitrarily typed exceptions is not composable, and is often lossy. It is already true that intermediate code should translate lower-level exception types to higher-level types as the exception moves across a semantic boundary where the error reporting type changes, so as to preserve correct semantic meaning at each level. In practice, however, programmers almost never do that consistently, and this has been observed in every language with typed propagated exceptions that I know of, including Java, C#, and Objective-C (see [Squires 2017] for entertaining discussion). This proposal actively supports and automates this existing best practice by embracing throwing values that can be propagated with less loss of specific information.

A `catch(type)` cascade is type-switching. We have learned to avoid switching on types, and now actively discourage that in our coding guidelines, yet a `catch(type)` cascade is often a form of type switch with all the problems of type switching.⁶ — In some cases, especially at lower levels, the code may intend to handle all errors. In that case, when the set of catchable errors can be statically known, automatic tools can often determine that we tested “all but one or two” of the possible values of a given error subtype, and so likely meant to test them all; and so when we add a new error value, we can get compile-time errors in code that tries to handle all errors but was not updated to handle the new error value (this will get even better when we get pattern matching). When throwing arbitrary types, the set of potential errors is open-ended (absent whole program analysis, which we can’t count on), and so it is much harder for tools to automatically detect a failure of intent to check all errors.

We’ve already actively been removing attempts to statically enumerate the types of arbitrarily-typed dynamic exceptions, in C++ and other modern languages. For example, C++ has already removed trying to enumerate

⁴ That is, “understand” to handle programmatically, more than just present them to a human such as by logging.

⁵ Michael Novak comments on how this aligns well with existing practices using dynamic exceptions: “I agree with this. One ‘patch’ for this problem that I find frequently employed (we do it ourselves in WIL [Windows Internal Libraries]) is simply normalization of the exception type. WIL strongly discourages creation of any custom exception types of your own. It normalizes on exactly one exception type and then provides helpers and macros to facilitate making it easy for everyone to throw that one type. This has worked very well for us as it allows us to bring along a set of relevant debugging information and normalize how errors are handled. I see this proposal overall being very similar (normalize on a single failure ‘exception’ type) and quite compatible with what we’ve been doing for a while now.”

⁶ And not avoidable in general because exception types can be arbitrarily unrelated. The usual advice to avoid a type switch is to replace it with a virtual function, which requires a tightly coupled class hierarchy that can be modified.

the types of dynamically typed exceptions in function signatures by removing `throw(type1, type2, type3)`, and instead made `noexcept` be untyped (just “can fail or can’t fail”). Similarly in Java, real-world uses of checked exception specifications quickly devolve to `throws Exception` which is equivalent to a typeless exception. Learning from the variety of error handling methods in Objective-C (described in a nutshell in the first 10 minutes of [Squires 2017]), Swift decided to pursue an always-untyped boolean `throws`, which is understandable but in my opinion overcorrects on the opposite side (and there are [ongoing recurring calls](#) to add a typed `throws` and/or a `Result` type to Swift, with sympathy from key participants [including Chris Lattner](#)). This proposal allows throwing rich values (not just boolean success/fail like C++ `noexcept` or Swift `throws`) that are able to represent all the same errors that types are used to represent in languages that throw typed exceptions (including today’s C++), by throwing a well-known common type rather than arbitrary types.

We’ve already standardized `std::error_code` and are moving to a world that is at least partly `std::error_code`-based. [P0824R1] makes this clear, pointing out for example that the actively-progressing `std::experimental::expected<T, E>` and `Boost.Outcome` are `error_code`-based and are expected to more aggressively embrace `error_code` (or an incremental evolution thereof) as their de-facto standard default code type. Our current status-quo path is to standardize these as library-only solutions, but if we continue down that path we will end up with a result that is still inferior because it’s missing language support for automatic propagation and to separate the normal and error paths cleanly.

Existing practice with error codes already tries to emulate the distinction between normal and error handling logic (using macros) to make the “normal” path clean. For example, the Windows Internal Libraries (WIL) designed its error-macro-handling system to enable error-code based code to emulate exception-based code by hiding all the branches that are about error handling. For example, real code commonly followed this structure:

```
// “Before” WIL error handling macros

HRESULT InitializeFromStorageFile(IStorageItem *pFile) {
    ComPtr<IStorageItem> spFileInternal;
    HRESULT hr = pFile->QueryInterface(IID_PPV_ARGS(&spFileInternal));
    if (SUCCEEDED(hr)) {
        hr = spFileInternal->GetShellItem(IID_PPV_ARGS(&m_spsi));
        if (SUCCEEDED(hr)) {
            hr = spFileInternal->get_CreationFlags(&m_sicf);
            if (SUCCEEDED(hr)) {
                hr = spFileInternal->get_FamilyName(&m_spszPackageFamilyName);
            }
        }
    }
    return hr;
}
```

So they “hid the error handling behind a wall of macros” as follows:

```
// “After,” using WIL error handling macros

HRESULT InitializeFromStorageFile(IStorageItem *pFile) {
    ComPtr<IStorageItem> spFileInternal;
    RETURN_IF_FAILED(pFile->QueryInterface(IID_PPV_ARGS(&spFileInternal)));
    RETURN_IF_FAILED(spFileInternal->GetShellItem(IID_PPV_ARGS(&m_spsi)));
    RETURN_IF_FAILED(spFileInternal->get_CreationFlags(&m_sicf));
}
```



```
    RETURN_IF_FAILED(spFileInternal->get_FamilyName(&m_spszPackageFamilyName));  
    return S_OK;  
}
```

This attempts to emulate (poorly, by hand and using macros) exception handling’s ability to distinguish the clean “success” path and propagate failures.

We *want* C++ programs to propagate errors automatically, and we *want* them to write distinct normal and error handling logic. It’s time to embrace this in the language, and effectively “[throw error_codes](#)” so that the language can (a) solve the performance problems with exceptions by throwing zero-overhead and deterministic exceptions, and also (b) make these new facilities easier to use robustly and correctly via language support for automatic propagation and [throw/catch](#) error paths that are distinct from normal control flow. — Then we can write future [filesystem](#)-like APIs naturally as

```
    directory_iterator& operator++() throws;    // operator + 1 function + 1 way to report errors
```

and the world will be a better place for everyone.

4.2 Proposed cleanup (separable, requires contracts or assertions)

SG Poll The 2018-04-11 SG14 telecon took a poll on pursuing this direction: [12-1-1 \(Favor-Neutral-Against\)](#). The one Against vote was on a previous draft of this section that proposed actually removing the existing throwing behavior, which would be a breaking change; that has now been demoted to “Future” below which addresses the concern.

“[With contracts,] 90-something% of the typical uses of exceptions in .NET and Java became preconditions. All of the `ArgumentNullException`, `ArgumentOutOfRangeException`, and related types and, more importantly, the manual checks and throws were gone.” — [Duffy 2016]

Even without this proposed extension, there is cleanup that we should do anyway in the standard library that would immediately benefit users. Work on the following direction is already in progress via [\[P0380R1\]](#) and [\[P0788R1\]](#) which in part systematically distinguishes between actual preconditions and recoverable errors, and [\[P0132R0\]](#) which aims to deal with `bad_alloc` in the standard library.

As noted in §1.1, preconditions, postconditions, and assertions are for identifying program bugs, they are never recoverable errors; violating them is always corruption, undefined behavior. Therefore they should never be reported via error reporting channels (regardless of whether exceptions, error codes, or another style is used). Instead, once we have contracts (expected in C++20), users should be taught to prefer expressing these as contracts, and we should consider using those also in the standard library.

The standard library should identify these cases, and aim to eventually replace all uses of exceptions and error codes for such bugs. [\[P0788R1\]](#) is a direct step in this direction, by moving toward systematically distinguishing preconditions (et al.) from recoverable errors throughout the standard library; it differs in that it does not yet recommend that standard components be required to use contracts.

Then, except for exceptions from user-defined types, the vast majority of standard library operations can be either `noexcept` or throw only `bad_alloc` (see also §4.3). [\[P0132R0\]](#) aims to offer suitable alternatives for code that aims to be hardened against heap exhaustion; with these in place we could consider reporting heap exhaustion differently from recoverable errors (but that is not proposed in this paper).

The following types are used (in part) for preconditions:

`logic_error` `domain_error` `invalid_argument` `length_error` `out_of_range` `future_error`

Note that there are uses that are not preconditions. For example, `future_error` is also used to report normal errors; also, `vector<T>::at` exists in order to throw an exception on a failed range check, and we may well want to keep that historical behavior.

For each of the above types:

- Add a non-normative note in the standard that we are maintaining the types’ names for historical reasons, but that these types should not be thrown to report precondition failures.
- In every place where the standard library is currently specified to throw one of the above types and *that upon review is or should be a precondition*, add an equivalent `[[expects: ...]]` contract or a [\[P0788R1\]](#) `Expects`: element.
 - **Future** (to avoid a breaking change now): Someday, remove the corresponding `Requires/Throws` clause (breaking change). If that eliminates the possibility of throwing, then additionally mark the function `noexcept`. ((And if §4.1 is adopted: Otherwise, mark the function `throws`..))

4.3 Proposed extension: Treat heap exhaustion specially

SG Poll The 2018-05-02 SG14 telecon took a poll on treating heap exhaustion as something distinct from other errors (i.e., not report it via an exception or error code) “in some way” (i.e., not tied to a particular method). The poll result was: 9-0-3-0-0 (SF-F-N-WA-SA).

4.3.1 The problem

“... if STL switched to making bugs [logic errors, domain errors] be contracts... and we could make `bad_alloc` fail fast, we could use the standard STL unmodified throughout Windows.”
 — Pavel Curtis, private communication

The following table summarizes the advice in §1.1 in cases ordered from least to most recoverable to resume successful program operation, and highlight a key question (middle row):

	What to use	Report-to handler	Handler species
1. Corruption of the abstract machine (e.g., stack overflow)	Terminate	User	Human
2. Programming bug (e.g., precondition violation)	Contracts, default to terminate	Programmer	Human
3. Heap exhaustion (OOM)	?	?	?
4. Recoverable error that can be handed programmatically (e.g., host not found)	Throw exception, or return error code	Calling code	Code
5. Alternate success (e.g., used smaller buffer, call me again for the next chunk)	Return success, this is not an error	Calling code	Code

However, a key question that repeatedly came up in private discussion was about case 3: Is heap exhaustion like all other errors, or should it be treated specially? The key differences are:

- Unlike case 1 (e.g., stack exhaustion), a heap allocation is always explicitly requested by a function in its source code, and so in principle the function or its caller could always test and attempt to respond to the failure.
- Unlike case 2, a heap allocation is not a programming bug because the programmer cannot in general test in advance whether it will fail before attempting the heap request.
- Unlike case 5, a heap allocation failure is not a form of success (by definition).

So the question boils down to whether to distinguish cases 3 and 4.

4.3.2 Discussion

Arguments against treating heap exhaustion (3) separately from other errors (4) include:

- **It is an explicit fallible request in source code.** Unlike stack exhaustion, heap exhaustion is always explicitly requested by a function, and so in principle the function or its caller could always recover.
- **Some (possibly lots) of code is correct today for heap exhaustion even without coding for it specially.** In a large call free when the millionth `push_back` fails, we can generally unwind and recover just fine today with just a `catch(...)` at the top.
- **Some alternatives exist.** For example, we could use the hook in §4.4 to let code that wants to handle allocation failure separately do so for static exceptions (not dynamic exceptions), at some nonzero cost.

Note That hook is at the cost of some overhead on all functions for which heap exhaustion is the only reportable error: not just the cost for the hook check itself in that function (which is expected to be cheap), but rather that it prevents the compiler from removing the error propagation logic entirely which we could do if the function were `noexcept`. Also, if we add the hook described there to the language primarily to use it in this way to treat heap exhaustion specially, then having that hook will incur small but nonzero overheads on *all* code that can throw.

Arguments in favor of treating heap exhaustion specially from other errors include:

- **Testing is much more difficult.** It is extraordinarily more difficult to test every failure point for heap exhaustion than to test every failure point for most other kinds of errors. Real-world code can be assumed to be not properly testing those failure points unless they are pervasively using technologies like fuzzers.
- **Recovery requires special care (aka “you can’t handle the truth”).** Most code that thinks it handles heap exhaustion is probably incorrect (see also previous point; there is some truth to the mantra “if you haven’t tested it, it doesn’t work”). Recovery from heap exhaustion is strictly more difficult than from other kinds of errors, because code that is correctly resilient to heap exhaustion can only use a restricted set of functions and language features. In particular, recovery code typically cannot ask for more memory, and so must avoid calling functions that could allocate memory, including functions that could fail by throwing another of today’s dynamic exceptions which would require increased allocation. Real-world systems that do handle heap exhaustion correctly are typically written to explicitly use `new(nothrow)` to opt into testing the result, and in many cases they already use functions such as those proposed by Ville Voutilainen in [P0132R0] that report allocation failures distinctly (e.g., `try_reserve` instead of just `reserve`).
- **In many programs, heap allocation failure can never happen as required by the standard (i.e., today’s C++ heap allocation semantics are unimplementable or unused).** As I wrote in [Sutter 2001]: (a) Reporting heap allocation failure is useless on systems that don’t commit memory until the memory is used, which the default on Linux with lazy commit mode 0 (see [LKA 2018], [Landley] Linux Memory FAQ, and [Gorman 2007] Chapter 13 and page 686’s annotated source for `vm_enough_memory()`), because on such systems `new` never actually fails (might as well be `noexcept`) and exhaustion actually manifests later as a hard fault on an ordinary operation that attempts to access the apparently-successfully-allocated memory (if the OOM killer selects this process); this is nonconforming, but it’s actual real-world behavior on such systems.⁷ (b) Allocation failure could never happen on virtual memory systems, because the actual symptom of nearing heap exhaustion is often increased thrashing

⁷ This is enabled by default in Linux, such as in Ubuntu and Red Hat Enterprise Linux. Some distributions disable it, [such as Android](#). These overcommit semantics are used in real-world production systems, such as at Google.

that slows the system more and more so that the program never reaches actual heap exhaustion at all. — If we re-specified allocation failures to result in `terminate` or be undefined behavior, that would appear to be an impossibly large breaking change on paper, yet such programs and environments would never notice the difference.

- **Reporting preference is often different.** For example, in [Koenig 1996] Andrew Koenig argued why halting programs on allocation failure is the best option in most situations, citing examples like the one in the previous bullet above.⁸ Treating allocation failure distinctly is also one of the two major motivations for §4.4 (see second bullet for a real-world example) which is an approach that could incur small but nonzero overheads on all code and might be more readily rejected if we treated heap exhaustion distinctly from other errors.
- **We have some precedent in the C++ standard for treating heap exhaustion differently from other errors.** See [syserr.syserr.overview]: “[Note: If an error represents an out-of-memory condition, implementations are encouraged to throw an exception object of type `bad_alloc` (21.6.3.1) rather than `system_error`. — end note.”
- **Contracts and heap exhaustion combined outnumber all other failure conditions by ~10:1.** For example, see [Duffy 2015]. Therefore, not reporting them as errors can greatly reduce the number of functions that can fail (emit exceptions or error codes). Corollaries:
 - **Most `std::` and user-written functions could be `noexcept`.** Once we ignore exceptions that should be preconditions, `bad_alloc` is the only exception that many standard library operations throw. The combination of changing standard library preconditions to contracts and treating heap exhaustion separately means that a large number of standard library and user functions could be made `noexcept`. In existing practice today, we have “exception-free” STL dialects that fail fast on `bad_alloc` as the basis for claiming they do not throw exceptions (unless thrown by user-defined functions types passed to that STL implementation).
 - **Enabling broad `noexcept` would improve efficiency and correctness (and `try-expression`, see §4.5.1).** Being able to mark many standard library and user functions as `noexcept` has two major benefits: (a) Better code generation, because the compiler does not have to generate any error handling data or logic, whether the heavier-weight overhead of today’s dynamic exceptions or the lightweight if-error-goto-handler of this proposal. (b) More robust and testable user code, because instead of examples like GotW #20 [Sutter 1997] where a 4-line function has 3 normal execution paths and 20 invisible exceptional execution paths, reducing the number of functions that can throw by 90% directly removes 90% of the invisible possible execution paths in all calling code, which is an important correctness improvement: All that calling code is more robust and understandable,⁹ and also more testable because its authors have fewer execution paths to cover. (In the future, it opens the door to entertaining default `noexcept`. Using `noexcept` more pervasively today also opens the door wider to entertaining a future C++ where `noexcept` is the default, which would enable broad improvements to optimization and code robustness.¹⁰)

⁸ Another example is that there may not be enough memory to even report a `bad_alloc`, such as if on the [Itanium ABI] the fallback buffer gets exhausted, or if on Windows there is insufficient pinnable stack space to store the exception as described in the Appendix. However, this paper’s static exceptions can resolve this particular issue on all platforms.

⁹ This also makes it more feasible to adopt §4.5 to make those exceptional paths visible, because it will remove 90% of the places to write a “this expression can fail” `try`-annotation.

¹⁰ Of course, changing the default would be a broad language breaking change and would need to be done on some backward-compatibility boundary. The point is just that it’s potentially desirable but probably not worth doing unless we could

- **Treating heap exhaustion the same as all other errors violates the zero-overhead principle by imposing overheads on all C++ programs.** As noted above, platforms and situations where heap allocation cannot be recovered from are common, but programs and programmers in that camp are paying for the current specification in performance across their programs. For example, using `noexcept` less often on functions that can only report `bad_alloc` incurs overheads on all programs as described in the previous bullet. As a specific example, resizing `vector<optional<T>>` is potentially slow only because of possible allocation failures. Titus Winters reports that one of the subtle things [Abseil] does that they are most happy with is to provide an explicit build flag for “does my default allocator throw” and to rely on “move constructors are assumed not to throw for anything other than allocation;” the result of that combination is that many move constructors can be made unambiguously `noexcept` and they measure better performance as a direct result ([example: abseil::optional](#)).

4.3.3 Proposal

This paper proposes the following basic approach, assuming rollout is done via a usual multi-step migration process to give code bases time to absorb the change as we do with most new features (i.e., deprecate and eventually remove, or provide compiler modes that make the new model off-by-default and later on-by-default and later required):

- Re-specify each language operation and standard library function that could throw `bad_alloc` today to have *implementation-specified* behavior, which each implementation specifies exactly one of the two following alternatives: (a) undefined behavior on systems that cannot detect heap exhaustion (e.g., Linux systems that overcommit); otherwise, (b) as today, call any installed `new_handler` else `terminate` by default, with the additional restriction that any installed `new_handler` must `terminate` (it can no longer `return` or `throw`).
- For each standard function for which allocation is the only reportable error, make the function `noexcept`. (We expect this to result in making a large majority of functions in the standard library `noexcept`, including default and user-replaced `::operator new`.)
- For code that wants to handle heap allocation failures, provide explicit “try to allocate” functions including the existing `new(nothrow)`¹¹ and `try_` functions like those being proposed in [P0132R0] (e.g., `bool vector<T,A>::try_reserve()`).

In early discussions, I have found that this fits actual existing practice much more pervasively than I expected, and in fact has been repeatedly reinvented:

- Code bases that use exception-free STLs already often terminate on allocation failure. (Example: Google’s production code uses compiler configuration that behaves as if every `new` were annotated `noexcept`, including the ones in `std::allocator`, as part of the strategy to enable using STL in an `-fno-exceptions` environment.)
- STL functions are already wrapped with `try_` variations in code bases that are robust to allocation failure. (Examples: Microsoft Word broadly uses wrapper functions around standard STL operations to add

get to a place where most functions are `noexcept`, and exceptions thrown for preconditions and `bad_alloc` are the two major things that stand in the way of that today for the standard library.

¹¹ Some implementations implement `new(nothrow)` in terms of throwing `new`. That’s fine inasmuch as they already need to make sure they do the right thing and at worst are just inefficient, but here is another motivating opportunity for them to switch to implement throwing `new` in terms of `new(nothrow)`.

functions such as `try_push_back`. Blackberry’s BB10 device running on QNX similar added `try_` STL operations. Making STL operations terminate on heap exhaustion would eliminate the primary reason these teams maintain nonstandard forks of their STL implementations.)

- It aligns with existing practice on systems like Linux where virtual memory overcommit means that our current C++ standard heap exhaustion design is fundamentally unimplementable and already ignored in practice, where `bad_alloc` can never happen and `new` is already de facto `noexcept`. (See previous subsection.)

Here are examples of what real-world code would do to adapt to the new model:

- **Program that never encounters heap exhaustion.** No action needed, won’t notice the change.
- **Program that doesn’t correctly handle `bad_alloc`.** The program can now always get a stack trace at the point of the allocation failure (and hook `new_handler` to launch the debugger or crash reporting tool if that is not the default behavior in their environment), instead of possibly unwinding the stack.
- **Program that relies on `bad_alloc` throws and does handle them.** Many systems typically already perform all allocation through a limited set of allocation functions, so they can just change the implementations of those existing functions (or wrap them) and implement them in terms of the `new(nothrow)` and/or `try_` functions and throw `bad_alloc` or some other exception on failure. Programs that have such a limited set of allocation functions can make this low-impact localized change to retain their current behavior.
- **Code that performs one big allocation, e.g., `image.load(“very_large.jpg”) or allocate(large_bufsize).`** Such code should use `new(nothrow)/try_reserve` explicitly and test failure locally. Code that can fall back to a smaller buffer size can just retry with the smaller size.
- **Microsoft Excel, File>Open `huge_spreadsheet.xlsx`: Big operation that causes lots of little-but-related allocations.** Today, 32-bit Excel can encounter heap exhaustion when enterprise users try to open huge spreadsheets. The status quo way it is handled is to display a dialog to the user, and for support to recommend using the 64-bit version of Excel. In this model, only a minimal change is needed to preserve the same behavior: Install a `new_handler` that invokes the existing failure dialog box display code. If the program wishes to continue if the whole large operation fails, it can make the allocations performed as part of the large operation use nonthrowing allocations (`new(nothrow)` and `try_` functions).
- **Microsoft Word.** Today, Word wraps STL with `try_` functions such as `try_push_back`. In this model, they could continue using their wrappers until they reach a convenient point to make the switch, then drop their wrappers and adopt the standard functions by making mechanical changes to use the standardized spellings of those `try_` operations wherever they are already used throughout the code (in most cases this will be a global find-and-replace).
- **Microsoft Office shared components.** Many shared components in Microsoft Office are using non-throwing user types with STL in `noexcept` contexts, with the knowledge that the only exceptions are heap exhaustion and contract violations. These components would be unchanged except their code would be slightly more efficient as the STL calls become `noexcept`.

SG Poll The 2018-05-02 SG14 telecon took a poll whether to pursue this approach to treating heap exhaustion distinctly from other errors. The poll result was: 2-6-3-2-0 (SF-F-N-WA-SA).

In summary, this paper proposes the following resolution:

	What to use	Report-to handler	Handler species
1. Corruption of the abstract machine (e.g., stack overflow)	Terminate	User	Human
2. Programming bug (e.g., precondition violation)	Contracts, default to terminate	Programmer	Human
3. Heap exhaustion (OOM)	<code>new(nothrow)</code> and <code>try_reserve</code> et al.	Special OOM-aware calling code	Code
4. Recoverable error that can be handed programmatically (e.g., host not found)	Throw exception, or return error code	General calling code	Code
5. Alternate success (e.g., used smaller buffer, call me again for the next chunk)	Return success, this is not an error	General calling code	Code

4.4 Optional extensibility hook: `set_error_propagation()`

Note This subsection is included because some users have requested the following functionality, especially if we do not adopt §4.3 to treat heap exhaustion separately in §4.3. If this is needed (e.g., because we didn’t adopt §4.3) and if it cannot be implemented in a totally zero-overhead way, then it could be a conditionally-supported feature.

We can additionally allow the program to register a global function to allow customizing what happens when exiting each function body with an error, intended to be used only by the owner of the whole program. This is useful for several purposes:

- **To integrate with an existing system’s error handling or logging policy.** Niall Douglas reports regarding a similar hook in his implementation: *“I’ve already had some big multinationals who are thrilled with this feature because they can encode the bespoke failure handling system they are already using into a custom policy, specifically custom logging and diagnostics capture.”*
- **To enable fail-fast throughout a system, even when invoking the standard library or third-party code.** For example, the layout engine for Microsoft’s [\[Edge\]](#) web browser ([EdgeHTML](#)) is designed to `terminate` if memory is ever exhausted. Today, EdgeHTML depends on a nonstandard fork of the standard library that does not throw exceptions and that terminates on `bad_alloc`. This is undesirable not only because it means the Visual C++ team receives requests to support an incompatible nonstandard variant of the standard library at least in-house, but because there is pressure to deliver the same products internally and externally and so there is pressure to document and ship this nonstandard mode which is undesirable for the community. Instead, with this hook EdgeHTML can accomplish its goal by using the standard STL, built in the mode described in §2 where all non-`noexcept(true)` functions are treated as `throws`, and installing a callback that calls `std::terminate()` on `ENOMEM` specifically and does nothing for all other errors. (But see also §4.3.)

Following the model of terminate handlers, we provide the ability register a callback, which is expected to invoke the previously installed callback:

```
using on_error_propagation = void (*)(std::error) noexcept;
atomic<on_error_propagation> __on_error_propagation = nullptr; // global variable
on_error_propagation set_on_error_propagation( on_error_propagation f ) {
    return __on_error_propagation.exchange(f);                // atomically set new hook
}
```

When exiting a `throws` function, the function epilog performs a jump to this common code, where `error e`, the `unwinding` flag, and the `return_address` are assumed to be already in registers:

```
// pseudocode
if (unwinding)
    [[unlikely]] if (auto x = __on_error_propagation.load(memory_order::relaxed)) x(e);
jmp return_address
```

Notes Overhead is expected to be minimal, but (importantly) non-zero. The expected code size cost is one unconditional `jmp` instruction in the epilog of a function declared `throws`: The return address will already be in a register, so just jump to the above common hook code (which will be hot in L1\$) which when finished jumps to the return address. The expected L1\$ overhead will be a constant 8-

16 bytes for the hook code above + 1 instruction per `throws` function. The expected L1D\$ overhead will be one pointer (the hook target).

For example, here is how to install a callback that will cause all heap exhaustion failures to fail-fast:

```
g_next_handler = set_on_error_propagation( [](error e){
    if (e == std::errc::ENOMEM) terminate();
    if (g_next_handler) g_next_handler(e);
    // else return == no-op, continue propagating
} );
```

4.5 Proposed extension (mostly separable): `try` and `catch` (addresses §3.1 group B)

SG Poll The 2018-04-11 SG14 telecon took a poll on pursuing these particular sugars: 5-1-8 (Favor-Neutral-Against). Some of the Against votes would like different sugars; I requested suggestions by email but have not received any yet as of this writing.

“The `try` keyword is completely redundant and so are the `{ }` brackets except where multiple statements are actually used in a try-block or a handler.” — [Stroustrup 1994]

“Failure to standardise this [operator `try`] means people may abuse `co_await` to achieve the same thing” — [P0779R0]

Notes Several reviewers felt strongly that this should be in the core proposal. For now and unless SG14 or EWG directs otherwise, I’m keeping it distinct; nothing in the core proposal depends on this.

4.5.1 `try` expressions and statements

Today, exceptional error handling flow is invisible by default; between the `throw` and the `catch`, all propagation is via whitespace. This makes it more difficult to reason about exceptional control flow and to write exception-safe code. One of the biggest strengths of `expected<T,E>` and `Boost.Outcome` is that they make intermediate error-handling control flow *visible* rather than invisible; but at the same time, they make it *manual* rather than automated. This section aims to make it both *visible* and *automated*, to help reduce today’s programmer mistakes and so that I don’t have to write articles like GotW #20 [Sutter 1997]. — See also related proposal [P0779R0].

This section proposes a **try-expression** and **try-statement**, where the keyword `try` can precede any full-expression or statement of which some subexpression could throw.¹² When applied to an expression, it applies to the maximal expression that follows (this `try` has lowest precedence). When applied to a statement, it is primarily useful for variable declarations that invoke fallible constructors.

For example:

```
string f() throws {
    if (flip_a_coin()) throw arithmetic_error::something;
    return try "xyzy"s + "plover";           // can grep for exception paths
    try string s("xyzy");                   // equivalent to above, just showing
    try return s + "plover";                 // the statement form as well
}

string g() throws { return try f() + "plugh"; } // can grep for exception paths
```

Notes This becomes even more attractive if §4.3 is adopted to dramatically reduce the number of potentially-throwing expressions and therefore the number of places one would write `try`.

¹² Swift, and the Microsoft-internal Midori project [Duffy 2016], also created a similar `try`. Unlike Swift, my interest is currently in plain `try` on an expression or statement, and not in Swift’s additional `try?` (monadic “auto-optional”) and `try!` (“auto-die”) variations.

[D0973R0] and other papers have proposed using `co_await` here instead of `try`, and with explicit support for `expected<T,E>` as an endorsed return type to be widely used.

In general I agree with the overall direction, namely: Just as the `expected<T,E>` proposal motivates this paper’s proposal to bring such a concept into the language as `throws`, I think that [D0973R0]’s suggestion to have an expression-level way to test the error result as `co_await` motivates this section’s proposal to bring such a feature into the language as `try` on expressions and statements. I think that is better than trying to shoehorn the feature into `co_await` which is not about error handling, and would leave error-handling code interspersed with normal code — in fact, trying to reuse `co_await` for this really is another example of trying to use normal control flow constructs for error handling, whereas error-handling paths should be distinct from normal control flow. See [P0779R0] section 1.2 for additional reasons why `co_await` is not a good idea here.

4.5.2 Compile-time enforcement, static guarantees

“Compile and link time enforcement of such rules is ideal in the sense that important classes of nasty run-time errors are completely eliminated (turned into compile time errors). Such enforcement does not carry any run-time costs – one would expect a program using such techniques to be smaller and run faster than the less safe traditional alternative of relying on run-time checking.” — [Koenig 1989]

If we did do this now, we have the opportunity to make `try` required in many cases that are immune from backward compatibility issues. For example, we can require that `try` must occur in the following cases on the described expression, on an enclosing expression, or on the entire statement in which the expression appears:

- In the body of a function declared with a static-exception-specification, any expression that contains a function call that is not `noexcept(true)`.
- In all code, any expression that contains a function call to a function declared with a static-exception-specification.

We can also make it an error to write `try` on an expression or statement when all subexpressions are `noexcept`.

For example:

```
try 1+2;           // error, cannot throw
return "xyzy"s + "plover"; // error, ""s and + could throw
try return "xyzy"s + "plover"; // ok, covers both ""s and +
return try "xyzy"s + "plover"; // same
return try ((try "xyzy"s) + "plover"); // ok, but redundant
```

Notes If we had a time machine, we could require `try` to precede every call to an expression that could throw. We cannot do that today without breaking backward compatibility.

For functions with static-exception-specifications we don’t need the time machine, because no such functions exist yet. In those functions and in calls to those functions, we can additionally require `try` to precede every expression some subexpression of which could throw, if we so desire.

Some languages, such as Midori, made it a static error to be able to leave function by throwing if it was marked as nonthrowing. We cannot do that with `noexcept` due to backward compatibility; but we can with the proposed `throws(false)` (which is otherwise a synonym for `noexcept`) and apply

these rules to give the static compile-time guarantee that such a function cannot exit by throwing. In such a function, the compiler never needs to generate the `noexcept` termination check.

In unconstrained generic code, when the concrete types are not known, it can be unknowable whether an expression throws. In that case, writing `try` can be permitted but not required, and if it is not written and in a given instantiation the expression can throw, the result is a compile-time diagnostic in the function template definition showing the offending type(s) and operations. Concepts can help with this, by stating what functions can throw.

4.5.3 catch sugars

We can also provide the following syntactic sugars for cleaner and clearer code that reduces ceremonial boilerplate without making the code too obscure or losing information:

- **“`catch{}`”**: To reduce ceremony and encourage efficient exception handling, we could consider letting `catch{ /*...*/ }` with no parameter be a convenience shorthand for `catch(error err){ /*...*/ }`.
- **“Standalone `catch`” / “less `try` where it’s unnecessary boilerplate (blocks)”**: To reduce the ceremony and nesting of `try{}` blocks, we could consider allowing standalone `catch` (not following an explicit `try`) as-if everything between the previous `{` scope brace and the `catch` were enclosed in a `try{}` block.¹³ After all, a major point of exception handling is to keep the normal path clean and direct, and today’s ceremony of `try{}` blocks and required extra nesting makes the normal path less clean and direct.

For example:

```
int main() {
    auto result = try g();           // can grep for exception paths
    cout << "success, result is: " << result;

    catch {                          // clean syntax for the efficient catch
        cout << "failed, error is: " << err.error();
    }
}
```

In my opinion, the combination of try-expressions/try-statements with this `catch` sugar is doubly desirable, because it lets us write clean, readable code that simultaneously avoids needless ceremony (e.g., the artificial scopes of `try{}` blocks) while adding back actually desirable information that was missing before (`try` to make the exceptional paths visible).

Finally, we could consider helping the common pattern where code wants to handle just one kind of error and propagate the rest. For example:

```
catch (error e) {
    if (e == codespace::something) {
        // do something different, else allow to propagate
    }
    else throw;
```

¹³ Note a potential danger: Anytime we consider omitting an explicit scope we run the risk of making the code feel (or actually be) unstructured and undisciplined. In this case, the preceding scope is already explicit in source code (everything from the scope-`{` to the `catch`); it’s certainly clear to the compiler, and in my opinion likely also clear to the programmer, and if that bears out with some usability testing then it would confirm that the `try{}` boilerplate really is only adding ceremony and nesting, not any actual information or clarity.

```
}
```

If we allowed naming an **error value** in the catch-handler, it would make this common scenario cleaner:

```
catch (codespace::something) {
    // do something different, else allow to propagate
}
```

4.5.4 Discussion

The “expression **try**” aligns well with hand-coded conventions already being adopted in the community, and brings them into the language. For example, it directly replaces the `OUTCOME_TRY` macro and enables writing the same naturally in the language in normal code:

[Douglas 2018] example	Possible future extension (not proposed)
<pre>outcome::result<int> str_multiply2(const string& s, int i) { OUTCOME_TRY (result, convert(s)); return result * i; }</pre>	<pre>int str_multiply2(const string& s, int i) throws { auto result = try convert(s); return result * i; }</pre>

This feature would help address the concern of some programmers (including the authors of `expected<T,E>`) who have expressed the desire to interleave normal and exceptional control flow in a way that today’s exception handling does not support well. Consider again the example from [\[P0323R3\]](#),

```
// P0323R3 expected<int,errc> style: as preferred by some
int caller2(int i, int j) {
    auto e = safe_divide(i, j);
    if (!e)
        switch (e.error().value()) {
            case arithmetic_errc::divide_by_zero: return 0;
            case arithmetic_errc::not_integer_division: return i / j; // ignore
            case arithmetic_errc::integer_divide_overflows: return INT_MIN;
            // Adding a new enum value causes a compiler warning here, forcing code to update.
        }
    return *e;
}
```

Using dynamic exceptions, the code can put the normal control flow together, but it creates a needless extra **try** scope in the normal path, and throwing types is brittle under maintenance if new failure modes are added:

```
// Today's C++ exception style: cleaner, but also more brittle (and more expensive)
int caller2(int i, int j) {
    try {
        return safe_divide(i, j);
    }
    catch(divide_by_zero) { return 0; }
    catch(not_integer_division) { return i / j; } // ignore
    catch(integer_divide_overflows) { return INT_MIN; }
    // Adding a new exception does not cause a compiler warning here, silently incorrect.
}
```

Using this section’s proposal, we could write a combination that arguably combines the benefits of both — note that the return value of `safe_divide` is now always a success result in the normal path and always a failure result in the `catch` path, never a conflated success-or-error result that must then separate its own code paths, yet the handling code’s structure is still essentially identical to the `expected<double,errc>` style:

P0323R3 example	Possible future extensions (not proposed)
<pre> expected<double, errc> caller(double i, double j, double k) { auto q = safe_divide(j, k); if (q) return i + *q; else return q; } </pre>	<pre> double caller(double i, double j, double k) throws { return i + try safe_divide(j, k); } </pre>
<pre> int caller2(int i, int j) { auto e = safe_divide(i, j); if (!e) { switch (e.error().value()) { case arithmetic_errc::divide_by_zero: return 0; case arithmetic_errc::not_integer_division: return i / j; // ignore case arithmetic_errc::integer_divide_overflows: return INT_MIN; } } return *e; } </pre>	<pre> int caller2(int i, int j) { try return safe_divide(i, j); catch { if (err == arithmetic_errc::divide_by_zero) return 0; if (err == arithmetic_errc::not_integer_division) return i / j; // ignore if (err == arithmetic_errc::integer_divide_overflows) return INT_MIN; } } </pre>

4.6 Q&A

4.6.1 Wouldn't it be better to try to make today's dynamic exception handling more efficient, *instead of* pursuing a different model?

“The unavoidable price of reliability is simplicity.” – C. A. R. Hoare

We can keep trying that too and continue hoping for a fundamental breakthrough, but we should not use that as a reason to delay investigating a zero-overhead-by-construction model.

We know that today's dynamic exception model has inherent overheads with no known solution for common and important cases such as memory-constrained and/or real-time systems, which require deterministic space and/or time cost of throwing an exception, with a statically computable upper bound on those costs. For space determinism (memory-constrained systems), I am not aware of any research progress in the past decade. For time determinism (real-time systems), proponents of today's dynamic exceptions expected suitability for real-time systems to be achieved before 2010; but, as noted in §2.3, there have been only a handful of research results (all over a decade old), notably [Gibbs 2005] which unfortunately is not suitable for general use because of its restrictions on the sizes of class hierarchies and reliance on static linking.

We cannot accept that “Standard C++ is not applicable for real-time systems” — that would be an admission of defeat in C++'s core mission as an efficient systems programming language. Therefore we know we cannot live with today's model unchanged.

The overheads described in §2.5 appear to be inherent in the *dynamic* and *non-local* design of today's dynamic exception model: In particular, today's model requires nonlocal properties and so there has to be nonlocal overhead in some form (implementations can compete creatively only on where to put it), and there is no known way to achieve space/time determinism for throwing. Therefore, even if we pour millions of dollars more into optimizing today's dynamic exception handling model, we know already that the best-case result would be that the overheads will still be present but somewhat lower (e.g., smaller static tables for less binary size bloat), and throwing will still not be deterministic in either space or time and so exceptions will still be unusable in real-time systems and/or systems without virtual memory.

4.6.2 But isn't it true that (a) dynamic exceptions are optimizable, and (b) there are known optimizations that just aren't being implemented?

“Almost anything is easier to get into than out of.” — Agnes Allen

Yes, but part (a) exposes a fundamental problem just in the way it's phrased, and part (b) could be telling.

The reason (a)'s phrasing exposes a fundamental problem is that it underscores that today's dynamic model *relies* on such optimizations, which is already a sign it fails to be true to C++'s zero-overhead spirit. Contrast:

- **“Pound of cure” [not C++'s ethos].** Today's “out-of-band dynamic exceptions” model is *modeled* as a dynamic feature, *specified* in a way that assumes dynamic overheads, then *relies* on optimization to optimize them away. Relying on the optimizer is a giveaway that we're talking about a “pound of cure” strategy, which is normally avoided by C++ and is a major reason why non-zero-overhead proposals like C++0x concepts have failed (they were defined in terms of concept maps, and relied on those being optimized away). That strategy is not appropriate for C++; it is the strategy of languages like Java and C# that

place other priorities first, and it is precisely because those languages rely on optimization to claw back performance losses that they can never be as efficient as C++ overall, not just in fact but also in principle.

- **“Ounce of prevention” [C++’s ethos].** This paper’s proposed “in-band alternate return value” model is *modeled* as a static (stack) feature, *specified* in a way that assumes strictly local stack/register use (designed to share the return channel under the covers), and so does not rely on optimizations to optimize overheads away. This kind of strategy is exactly why and how C++ has been uniquely successful as a zero-overhead language: because it (nearly always) avoids incurring overheads in the first place, by construction. Furthermore, it doubles down on C++’s core strength of efficient value semantics.

Could we try to add requirements to today’s dynamic exception model, to say things like “local cases must be as efficient as a forward `goto`”? Possibly, but: We have never specified such a thing in the standard, and in this case it would address the problem only in degree, not in kind — it would be a “mandatory optimization” (“required poundage of cure”) rather than a correction to fundamentally fix the specification of the operation as a static and local feature, instead of as a dynamic and non-local feature.

For an example of (b), see [Glisse 2013]. The intention is to short-circuit `throw` and `catch` when that is visible to the optimizer (possibly across inlining boundaries), and it could result in performance gains. That it has been mostly ignored and not viewed as a priority is arguably telling. Granted, there could be many reasons why it’s not fixed, such as Clang being maintained in part by organizations that themselves ban exceptions and who therefore are disincentivized to optimize them; but if so then the maintainers’ being uninterested because they already abandoned exceptions outright is data too.

Note Gor Nishanov mentions a specific potential optimization of today’s dynamic exception handling model that could alleviate at least the need to have `filesystem`-like dual APIs. Gor writes:

Consider the case where `catch` and `throw` are in the same function and handler does not call any functions that compiler cannot see the bodies of (to make sure there is no sneaky calls to `current_exception` or `throw`; hiding inside) and the handler does not rethrow. In that case, `throw <expr>;` can place the exception object on a stack and then does a “normal” `goto` into the handler.

With that rather straightforward optimization we no longer need to have duplicate APIs as in `<filesystem>`. In our implementation, all the dual APIs are implemented as:

```
fs_xxx(bla, ec) { ec = fs_xxx_impl(bla); }
fs_xxx(bla)    { if (auto ec = fs_xxx_impl(bla)) throw ec; }
```

with the exception that we don’t throw `ec`, but a fatter object.

If [a caller] does local `catching` and only uses ‘throwing’ version of the API, with that simple optimization we will have the codegen identical to a version that request the `ec` and then does handing of the `ec` with an `if` statement. True, the [caller’s] `catch` syntax would be more verbose, but, that is something that can be addressed in either exception model.

4.6.3 Can I catch **error** values and dynamic exceptions?

Yes, just write `catch(...)` to catch both or write multiple `catch` blocks for different thrown types as today.

For example:

```
int f() throws { throw std::errc::ENOMEM; } // report failure as ENOMEM
int g() { throw std::bad_alloc; }          // report failure as bad_alloc
int main() {
    try {
        auto result = f() + g();
    } catch(error err) {                    // catch 'error'
        /*...*/
    } catch(std::exception const& err) {    // catch 'std::exception'
        /*...*/
    }
    try {
        auto result = g() + h();
    } catch(...) {
        /*...*/
    }
}
```

To invoke the translation to a common error type, use a function (possibly a lambda):

```
int h() throws { // bad_alloc → ENOMEM
    return f() + g();
}
int main() {
    try {
        auto result = h();
    } catch(error err) { // catch 'error', incl. translated 'std::bad_alloc'
        /*...*/
    }
    try { []() throws { // bad_alloc → ENOMEM
        auto result = f() + g();
    }(); }
    catch(error err) { // catch 'error', incl. translated 'std::bad_alloc'
        /*...*/
    }
}
```

4.6.4 Can `error` carry all the information we want and still be trivially relocatable? For example, `filesystem_error` contains more information, such as source and destination paths

Yes, you can have a `filesystem_error` type that includes additional information and pass it losslessly via `error`. Note that “trivially relocatable” is a surprisingly loose requirement; it is satisfied by many `std::` types, including containers and smart pointers.

4.6.5 What about allowing the function to specify the type it throws (e.g., `throws{E}`)?

SG Poll The 2018-04-11 SG14 telecon took a poll on pursuing this direction: 4-2-5 (Favor-Neutral-Against).

“We want to be able to pass arbitrary, user-defined information ... to the point where it is caught. Two suggestions have been made for C++: [1] that an exception be a class, and [2] that an exception be an object. We propose a combination of both.” — [Koenig 1989]

“[In Midori] Exceptions thrown by a function became part of its signature, just as parameters and return values are.” — [Duffy 2016]

Yes, we can add that extension later if experience shows the need. This paper currently does not propose this because it was more controversial in SG14, but this subsection captures its motivation and design. If we were to adopt this feature, we would reconcile the syntax with conditional `throws` (see §4.1.4) so that they do not collide.

A possible extension is to allow optionally specifying a custom error code type that can carry additional information until it converts to `error` or another thrown type. There are two main motivations:

- To return a **bigger** or **richer** error type, for example, one that directly carries additional information without type erasure. Because `std::error` can already represent arbitrarily rich information by wrapping an `exception_ptr`, and any performance advantage to doing something different would be expected to be in exceptional (throwing) cases only where optimization is rarely useful, justifying this motivation would require providing examples that demonstrate that throwing an alternative type gives better usability (at `catch` sites) and/or performance by avoiding type erasure.
- To return a **smaller** error type. For example, to return an error type that is smaller than two pointers. Because empirical testing shows that on common platforms there is little measurable performance difference in return-by-value of trivially copyable values of size from 1 to 32 bytes, justifying this motivation would require providing examples of platforms that benefit from throwing a smaller type.

The idea would additionally permit a function to declare a type-specific **static-exception-specification** of `throws{E}` (or other syntax, just using `{}` for now for convenient discussion without collision with `throws(cond)`) which uses the type `E` instead of `error` and otherwise behaves the same as in the previous section. In this model, a specification of `throws` is syntactic sugar for `throws{error}`. For example:

```
string f() throws{arithmetic_error} {
    if (flip_a_coin()) throw arithmetic_error::something;
    return "xyzzys" + "plover";    // any dynamic exception is translated to arithmetic_error
}
```



```
string g() throws { return f() + "plugh"; }
// any dynamic exception or arithmetic_error is translated to error
```

Or, `std::filesystem` might want to say:

```
directory_iterator& operator++() throws{filesystem_error};
```

For a function declared with a static-exception-specification, if it is declared `throws{E}` then additionally:

- `E` must be a single type name and must be default-constructible, noexcept-movable,¹⁴ and convertible to and from `error`. If `E` is not relocatable, then the implementation needs to call destructors of moved-from `E` objects when returning.
- All the rules in §4.1.1 apply as written, except replacing each mention of `error` with `E`. — Corollary: When a function declared `throws{E1}` calls another declared `throws{E2}` which throws an exception that is not handled, the `E2` object is converted to `E1`; if there is no conversion, then the `E2` object is converted first to `error` and then to `E1`.

The potential benefits of this extension include:

- It enables reporting rich errors while also encouraging handling them locally within a subsystem.
- Experience with Expected [P0323R3] and Midori [Duffy 2016] indicates that being able to handle rich errors locally is desirable and useful, while still propagating more generic errors distantly.

Note Some reviewers asked whether we would be able to evolve the `error` type over time. I believe this would enable that, by allowing more than a single well-known type which could be used for future evolutions of (or alternatives to) `error` itself. But I don't think that's sufficient motivation in itself to allow `throws{E}`, as we believe we have enough experience with `error_code` and its evolutions to specify `error` well to be future-proof for decades. The main reasons to allow `throws{E}` is to satisfy the requirements of code that uses Expected today with an error type other than `std::error_code`.

If we pursue this optional generalization, we should address the following concerns:

- Show use cases that demand specifying a custom type `E: error` as described herein is efficient, and sufficiently flexible to represent all errors in STL, POSIX, Windows, etc., and can wrap arbitrary richer types and transport them with trivial operations. What uses cases are not covered? Simple beats complex, unless the complexity is essential (to expose an important semantic or performance distinction so that the programmer can control it).
- Address how to discourage large error types: These objects should remain small and avoid heap allocation. One AVX512 register or one cache line is ideal.
- Demonstrate actual benefits from a smaller error type: A smaller type would unlikely give measurable savings over returning a two-pointer-sized `error`. The [Itanium ABI] already optimizes that use case. Niall Douglas reports that when benchmarking options for Boost.Outcome, he found no statistically significant difference for returning up to 32 bytes of data as compared to anything less for an Ivy Bridge CPU, although the answer may be different on other architectures.

¹⁴ Trivially relocatable is encouraged, for efficiency so that it is easier for platform ABIs to return in registers. However, this mechanism does not rely on it, and because the choice of `E` is exposed and left to the function author, it meets the zero-overhead principle of getting only the overheads the author opts into.

4.6.6 How will `vector<throwing_type>` work?

The existing `std::vector` works the same as today. If `std::vector` is instantiated with a type that declares a function with a static-exception-specification, then any conditional `noexcept` or `*noexcept*/nothrow*` trait behaves as if the function were declared `noexcept(false)`.

Looking forward, if there is a `std2` where we can change existing entities, then personally I would recommend that `vector` simply mark any function that might throw as `throws`:

- It's simple to teach and learn: A user of `vector` would just write error handling code without worrying about whether for a particular vector specialization error might actually happen or not (as such code has to do today anyway if it is generic code that uses a `vector<T>` for arbitrary `T`).
- It's also efficient: Because `throws` is efficient, there is no longer a performance incentive to perform a conditional calculation to see if for a given specialization it might be able to guarantee it doesn't throw, in order to suppress the exception handling machinery overhead.

4.6.7 What about `move_if_noexcept()`? conditional `noexcept`? traits?

They all work the same as today and react as if the function were declared `noexcept(false)`. This proposal is not a breaking change.

4.6.8 Will we want more `type_traits` to inspect dynamic vs. static exceptions?

I don't think so. The existing ones, and the `noexcept` operator, continue to work as designed. If anything, I think this proposal's effect on the exception-related type traits will be just to use them less often, not to add new ones, since many uses of the traits are motivated by wanting to elide the overhead of today's exception handling (notably to write conditional `noexcept` tests).

4.6.9 But people who use exceptions typically can't use dynamic memory anyway, right? So there's no benefit in trying to help them use exceptions on their own.

No. That may be true of some environments, but not of most. For example, the Windows kernel allows dynamic memory allocation (after all, it owns and manages the heap) but currently bans exceptions.

4.6.10 How does the `try`-expression in §4.5 compose with `co_await`?

§4.5 suggests allowing `try` on any statement or expression (including subexpressions) that could throw. In [O'Dwyer 2018], Arthur O'Dwyer makes the excellent point that "coloring" like `try` and `co_await` should be composable, and used only for essential qualities. His example is:

```
auto result = try (co_await bar()).value();
```

Recall this similar example from §4.5, which proposes allowing `try` on an expression or statement any sub-expression of which can throw, and indicates that mental model implies the following two lines are equivalent:

```
try return "xyzyz"s + "plover";           // ok, covers both ""s and +
return try "xyzyz"s + "plover";           // same
```

For the same reasons, the following are also equivalent ways to call an asynchronous function that can fail:

```
int result = co_await try foo();
int result = try co_await foo();
```

Incidentally, this is a counterexample that argues against the suggestion in [D0973R0] to co-opt `co_await` to mean what §4.5 suggests be a `try`-expression: If `co_await` can only mean one thing at a time, and one abuses it so that sometimes that thing is error handling, then how do you express invoking an operation that both can report an error and is asynchronous? (There are many other reasons `co_await` should not be co-opted for this meaning, including that error handling should be distinct from all normal control flow.)

4.6.11 Wouldn't it be good to coordinate this ABI extension with C (WG14)?

Yes, and that is in progress.

This paper proposes extending the C++ ABI with essentially an extended calling convention. Review feedback has pointed out that we have an even broader opportunity here to do something that helps C callers of C++ code, and helps C/C++ compatibility, if C were to pursue a compatible extension. One result would be the ability to throw exceptions from $C++ \rightarrow C \rightarrow C++$ while being type-accurate (the exception's type is preserved) and correct (C code can respond correctly because it understands it is an error, even if it may not understand the error's type).

It could simplify our C++ implementation engineering if C functions could gain the ability to return A-or-B values, so for example:

```
_Either(int, std_error) somefunc(int a) {
    return a > 5 ? _Expected(a) : _Unexpected(a);
}
// ...
_Either(int, std_error) ret = somefunc(a);
if(ret)
    printf("%d\n", ret.expected);
else
    printf("%f\n", ret.unexpected);
```

Here `_Either` would be pure compiler magic, like `va_list`, not a type. Under the hood, functions would return with some CPU flag (e.g., carry) set if it were a right value, clear if it were a left value. An alternative is setting a bit in the thread environment block which probably is much better, as it avoids a calling convention break, though potentially at the cost of using thread local storage. It would be up to the platform vendor what to choose (this is an ABI extension), but the key is that C and platforms' C-level calling conventions would be extended to understand the proposed union-like return of values from functions to use the same return channel storage, with a bit to say what the storage means.

In C++, functions marked `throws` would, in C terms, return `_Either(T, std::error)` and this is how the exception throw would be unwound up the stack, by the C++ compiler testing after every `throws` function call if an unexpected `std::error` value was returned, and if so returning that `std::error` up to caller.

Thus, these C++ functions:

```
extern "C++" int do_something(double) throws;
extern "C++" double do_something_else(int) noexcept;
```

would in C become:

```
extern _Either(int, std_error) _Z3do_somethingd(double);  
extern double _Z3do_something_elsei(int);
```

A major benefit of (also) doing it at the C level is that C is the lingua franca portable platform language that all other languages bind to. Therefore this facility would help integrate C not only with C++ error handling, but also with Rust, Swift, and many other languages that can speak C and also return A-or-B from functions as their main error handling implementation. The carrot for WG14 would be that C facility would benefit all C speaking languages, and help C be more useful in its key role as a glue language. A notable gain is that C could now receive exception throws from C++, and propagate them, perhaps even convert them for other languages.

Further, there are benefits for embedded use of C:

- On many common microcontrollers, this allows a more efficient implementation of any function that currently produces/propagates errors using return plus an out parameter for the return value. Adding parameters to a function is expensive, more expensive than using a flag.
- This provides a usable replacement for `errno` which is not an option in interrupt service routines or preemptive run-to-completion kernels (either adds ISR latency or potentially yields false results).

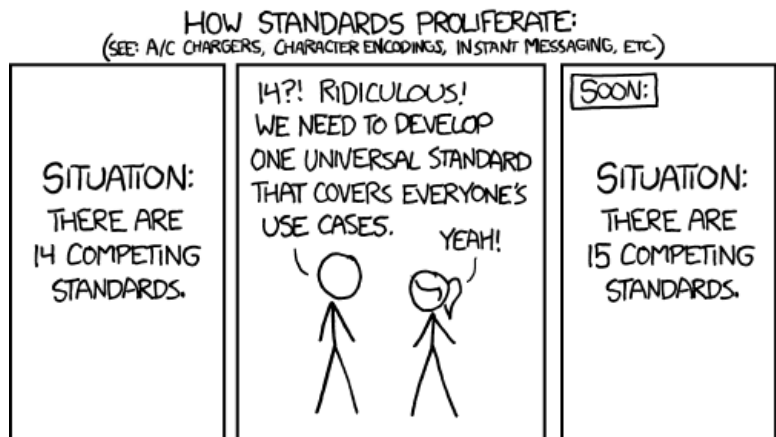
In C++, I would like to see group discussion of this question and get direction of SG14 and EWG regarding their interest in this proposal if C does, or does not, also make a coordinated ABI call convention extension with C++. Is coordination with C necessary for this proposal to be interesting for C++, or should this proposal be explored as of potential interest for a future C++ regardless of what C does?

5 Dangers, and “what to learn in a TS”

“Adding a mechanism to ‘solve the error-signaling problem’ is more likely to add yet-another-alternative to this mess, than to solve it.” — [P0976]

Here are two major groups of dangers we must address (e.g., in a TS) before considering adopting this proposal into a draft standard, with sample hypotheses that if validated would address them.

(1) Ending up with N+1 alternatives. To avoid [xkcd 927](#) (see right), we must validate that this proposal is unlikely to increase the number of alternative C++ error handling methods. One way to do that is to demonstrate it can replace at least one of the existing alternatives already in development and use (e.g., Outcome, Expected), for example if the proposers of one of those documents that this proposal can be viewed as a direct generalization that can subsume the existing alternative. That would demonstrate that this proposal at least does not increase the number of alternatives.



(2) Replacing known problems with something having unknown problems. We must create an experimental implementation and try it against real projects to validate its efficiency and usability, and to identify its currently-unknown problems. Also, we reduce this risk by documenting how the proposal directly relates to existing practice with known qualities in C++ and other languages. Note two specific cases of the latter problem:

- **The danger that this proposal might encourage excessive local propagation (e.g., a pattern of writing otherwise-unnecessary code to `catch` from a `throws{E}` before `E` is converted to `error`).** We must validate that projects using an experimental implementation still throw and catch exceptions at whatever distance is normal, without resorting to otherwise-undesirable local catches before a more specific error type is translated to `error`.
- **The danger that we cause users to reinvent a form of RTTI.** We must validate that projects do not use these facilities to catch `errors` at higher code levels in a way where they then perform run-time type/categorization efforts that are tantamount to duplicating RTTI in a way that falls into today’s problems of RTTI (see Note in §2.5).

Finally, we must not mimic other languages’ choices just because things appeared to have worked out well for them. Languages are different, even languages as similar as C and C++, and what works well in one language does not necessarily work well or at all in another. So, although the rationale and core design of this proposal arrives at similar conclusions and basic designs as existing practice in C++ (e.g., `std::error_code`, `Expected`, `Outcome`) and other modern languages (e.g., Midori, and in parts CLU, Haskell, Rust, and Swift), and so benefits from experience in those languages, it is not motivated by mimicry and we cannot rely only on experience in those languages. This is a proposed feature for C++, and it has to be prototyped and tested in a C++ compiler and with C++ code to get relevant data about its actual performance and usability.

“Haskell does something even cooler and gives the illusion of exception handling while still using error values and local control flow” — [Duffy 2016]

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Appendix: Illustrating “stack” exception allocation (Windows)

Below is a simple test program to measure the stack overhead of exception handling using the Windows model. Recall from §2.5.1, point (2), that the Windows exception handling model attempts to avoid having to heap-allocate exception objects by optimizing them to be physically on the stack, but not with usual stack semantics.

This test program has only two short functions, one of which calls the other three times. The test measures the stack impact of:

- “Current” vs. “Proposed”: reporting errors via exceptions (“Current”) vs. a `union{result; error;} + bool` (“Proposed”, a naïve implementation of this proposal written by hand; if this proposal were implemented, the source code for both tests would be the same, `try/catch`).
- Frame-based vs. table-based: in release builds for x86 (no tables) and x64 (table-based).

The code follows at the end. The following prints the memory offsets of various local and exception/error objects relative to a local stack variable in the topmost caller. The runs were on Release x86 and Release x64 builds using Visual Studio 2017 15.7, which use frame-based and table-based implementations, respectively, of today’s dynamic exception handling.

Release x86 (frame-based)			Release x64 (table-based)		
	---- Stack offsets ----			---- Stack offsets ----	
	Current	Proposed		Current	Proposed
test.c:	-56	1	test.c:	-96	1
test2.c:	-120	2	test2.c:	-216	2
test2.exception:	-139	-51	test2.exception:	-240	-72
test.exception:	-139	-67	test.exception:	-240	-96
test2.c:	-2,280	3	test2.c:	-19,000	3
test2.exception:	-2,299	-35	test2.exception:	-19,024	-48
test.exception2:	-2,299	-67	test.exception2:	-19,024	-96
test2.c:	-4,424	4	test2.c:	-37,704	4
test2.exception:	-4,443	-19	test2.exception:	-37,728	-24
test.exception3:	-4,443	-67	test.exception3:	-37,728	-96
main.exception:	-139	5	main.exception:	-240	8

The offsets show how in the “Current” (today’s exception handling) case, not only are the stacks fattened up by the `try/catch` machinery in both x86 and x64, but the effect mentioned in §2.5.1 indeed ‘pins’ the already-destroyed-and-otherwise-ready-to-reclaim stack while each exception is handled. The mockup of this proposal flattens the stack, including showing reuse of the same storage for `test.exception`, `test.exception2`, and `test.exception3` (here by hand coding, but I expect existing optimizers to routinely do it if the two lines marked `// can reuse result` were preceded with `auto` to declare new nested variables, and we weren’t taking their addresses to print them).

Note On non-Windows platforms, using gcc ([Wandbox](#)) and Clang ([Wandbox](#)), the three exceptions similarly create three distinct allocations, but on the heap with addresses far from the stack. Still, they are required by the current model to be three distinct addresses there too, and three distinct heap allocations modulo heroic optimizations.

Finally, note that this measures only stack space savings on Windows platforms, and does not attempt to measure the other primary savings this proposal aims to achieve (e.g., elimination of global state such as tables).

Sample code

```
#include <vector>
#include <string>
#include <iostream>
#include <iomanip>
using namespace std;

//=====
//    Helpers and instrumentation

//    Quick-and-dirty GSL (avoiding assert because I'm testing only release builds)
namespace gsl {
    void Expects(bool b) { if (!b) abort(); }
    using index = ptrdiff_t;
}

//    Counters
vector<string> labels;
vector<int> run1, run2;
vector<int>* prun = nullptr;
char* pbase = nullptr;

//    Record the address of a given object relative to the current test stack base
template <class T>
void addr(T& p, const char* msg) {
    gsl::Expects(prun != nullptr && (prun == &run1 || prun == &run2));
    if (prun == &run1) labels.push_back(msg);
    else gsl::Expects(labels[run2.size()] == msg);
    prun->push_back(&(char&)p - pbase);
}

//    Print results
void print_results() {
    gsl::Expects(labels.size() == run1.size());
    gsl::Expects(run1.size() == run2.size());

    cout.imbue(std::locale(""));
    cout << setw(45) << "---- Stack offsets ----" << endl;
    cout << setw(34) << "Current";
    cout << setw(11) << "Proposed" << endl;

    for (gsl::index i = 0; i < (gsl::index)labels.size(); ++i) {
        cout << setw(20) << labels[i] << ": ";
        cout << setw(11) << run1[i];
        cout << setw(11) << run2[i] << endl;
    }
}

//=====
//    Test types

struct success { int* _; int32_t __; };
struct error { int* _; int32_t __; };
struct folded { union { success s; error e; } u; bool b; };
```

```
//=====
//    Test for today's EH

namespace Current {

    success test2() {
        char c;
        addr(c, "test2.c");

        try {
            throw exception();
        }
        catch (exception& e)
        {
            addr(e, "test2.exception");
            throw;
        }

        return success();
    }

    success test() {
        char c;
        addr(c, "test.c");
        success result;
        try {
            result = test2();
        }
        catch (exception& e)
        {
            addr(e, "test.exception");
            try {
                result = test2();
            }
            catch (exception& e)
            {
                addr(e, "test.exception2");
                try {
                    result = test2();
                }
                catch (exception& e)
                {
                    addr(e, "test.exception3");
                }
            }
            throw;
        }

        return result;
    }

    int main()
    {
        prun = &run1;

        char base;
        pbase = &base;

        try {
```

```

        auto result = test();
        return result._ != nullptr;
    }
    catch (exception& e)
    {
        addr(e, "main.exception");
        return 0;
    }
}

//=====
//      Test for proposed EH

namespace Proposed {

    folded test2() {
        char c;
        addr(c, "test2.c");
        folded result;

        //try {
            result.u.e = error();
            result.b = false;
        //}
        //catch (exception& e)
        //{
            addr(result.u.e, "test2.exception");
            return result;
        //}

        result.u.s = success();
        result.b = true;
        return result;
    }

    folded test() {
        char c;
        addr(c, "test.c");
        folded result;
        //try {
            result = test2();
        //}
        //catch (exception& e)
        if (!result.b)
        {
            addr(result.u.e, "test.exception");
            //try {
                result = test2();           // can reuse 'result'
            //}
            //catch (exception& e)
            if (!result.b)
            {
                addr(result.u.e, "test.exception2");
                //try {
                    result = test2();       // can reuse 'result'
                //}
                //catch (exception& e)
                if (!result.b)

```

```

        {
            addr(result.u.e, "test.exception3");
        }
    }

    return result;
}

int main()
{
    prun = &run2;

    char base;
    pbase = &base;

    //try {
        auto result = test();
        if (result.b) return result.b == true;
    //}
    //catch (exception& e)
    if (!result.b)
    {
        addr(result.u.e, "main.exception");
        return 0;
    }
}

//=====

int main() {
    Current::main();
    Proposed::main();
    print_results();
}

```